



Vortex induced vibration analysis of a twin-box bridge deck by means of 3D LES simulations

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ARTICLE INFO

Keywords:

3D LES
Twin-box
Vortex shedding
Vortex induced vibrations
Force coefficients
Strouhal number
Pressure coefficient distribution
Correlation
Contribution
Work

ABSTRACT

Twin-box decks are prone to suffer vortex-induced vibrations (VIV). Although this phenomenon has been widely studied experimentally, there are still gaps in our understanding about the complex interplay between the incoming flow, the windward and leeward boxes, and the potential oscillation of the deck. This work exploits the ability of 3D LES simulations to simulate complex aeroelastic phenomena to delve into the linkage between aerodynamic forcing and heave oscillation through detailed analysis of comprehensive datasets. Several reduced velocities are studied in the VIV-prone range of the bare deck of the Stonecutters Bridge, analysing the spectra of the time-histories of force coefficients, for the overall deck and individual boxes, and heave oscillations. Similarly, the contribution of the local lift coefficient has been studied along with the work done by the deck and the individual boxes at different reduced velocities. It has been found that the leeward box is the one governing the overall VIV response of the deck. The spanwise-averaged time-averaged work distribution around the deck permits the identification of those regions contributing to the heave oscillation build-up, enabling the design of aerodynamic countermeasures for mitigation of VIV tailored for the specific deck geometry and dynamical properties of the deck.

1. Introduction

The development of multi-box bridge decks has enabled the design and construction of even longer cable-supported bridges, providing improved performance against flutter instability, as the works conducted by Qin et al. (2007), Larsen et al. (2008), Kwok et al. (2012), Chen et al. (2014) or Yang et al. (2015) have confirmed. Some examples of twin-box bridges already in service are the Stonecutters and the Xihoumen bridges, both in China, the Yi Sun-sin Bridge, in South Korea and the Çanakkale Bridge in Turkey. Although these bridges can withstand very high wind speeds, they are prone to suffer vortex induced vibrations (VIV) as the box girder located downwind is immersed in the wake of the upwind one. Vortices shed from the upwind box, may impinge upon the downwind box, generating periodic forces capable of exciting the deck. The VIV phenomenon typically takes place at low wind speeds, and it is characterised by self-sustained and self-limited oscillations, which seldom surpass the amplitude of half the depth of the deck (Simiu and Scanlan, 1986). Although these oscillations are not catastrophic, they may cause both serviceability and fatigue related issues.

Early studies on the VIV of twin-box decks were related with the Stonecutters Bridge and the Xihoumen Bridge projects. In Larose et al.

(2003), the VIV responses of the Stonecutters Bridge bare deck configuration, and also considering two sets of aerodynamic appendages, were studied at Reynolds numbers one order of magnitude higher than in standard wind tunnel tests. In Larsen and Poulin (2005) and Larsen et al. (2008), the vortex formation mechanism and the Reynolds number effects on the guide vanes performance were analysed in detail for the same bridge. In Ge et al. (2011), several aerodynamic countermeasures such as guide vanes, grid plates and adjustable wind barriers were studied for the mitigation of the VIV response of the Xihoumen Bridge.

More fundamental studies were addressed by Laima et al. (2013) for the Xihoumen deck cross-section. In this reference, vortex shedding and VIV responses were studied experimentally, but also adopting forced-oscillation 3D LES simulations for providing a qualitative description of the flow around the deck. This was an early application of numerical methods in the study of the VIV of twin-box decks, and will be further commented upon when referring to numerical studies. In Li et al. (2014), the Reynolds number effects on the pressure distribution, aerodynamic force, vortex shedding frequency and VIV response were studied by means of wind tunnel tests. In Chen et al. (2014) a wind tunnel parametric study considering four different gap ratios

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was reported, focusing on the flow features obtained using the Particle Image Velocimetry (PIV) system. [Laima and Li \(2015\)](#) were the firsts to analyse an ample range of gap widths for a twin-box bridge, by means of wind tunnel tests, considering the static and dynamic responses, finding a critical value for the gap to depth ratio for which a sudden change in the flow pattern takes place. Similarly, four VIV types were described based on the gap distance.

In recent years, experimental studies have been completed for decks with three-box girders, such as [Yang et al. \(2021\)](#), where the vortex-excited force was analysed based on the synchronous measurement of pressures and model displacements, providing the distribution of the mean and fluctuating vortex-excited force and the contribution. In [Wang et al. \(2022\)](#), another three-box deck was analysed, considering in this case the distribution of the aerodynamic work at VIV. In this last reference, 2D CFD simulations for the static deck were reported for the qualitative description of the flow, depending on the presence or absence of grid plates proposed as a mitigation device for VIV excitation. In [Xu et al. \(2023\)](#), wind tunnel tests were used to study the VIV response of twin-box decks considering different gap distances, focusing on the mean and fluctuating pressure coefficients, time-frequency characteristics of the aerodynamic force and the contribution values of aerodynamic forces. The authors proposed the use of skirt plates and wind fairings or winglets to mitigate VIV. The effect of these countermeasures was shown by the modification of the contribution values of the distributed aerodynamic forces. [Chen et al. \(2024\)](#) obtained the fluctuating pressure coefficient distributions at different reduced velocities and angles of attack, evaluating also the correlation between distributed and overall vortex-induced force. In this last reference, modal decomposition techniques were applied for the fluctuating pressure fields.

Published works addressing VIV by means of CFD simulations are scarce, mainly due to the limitations in accuracy, and very large computing time demands ([Marra et al., 2011](#)). Among the works studying VIV by means of CFD approaches, [Sarwar and Ishihara \(2010\)](#) studied the maximum amplitude response in heave oscillation of a box girder considering different aerodynamic countermeasures using 3D LES simulations. In [Li et al. \(2022\)](#), the VIV response of a single Π -shaped girder was studied by means of wind tunnel tests and CFD analysis, finding a qualitative agreement for the VIV response between 2D URANS simulations and experiments, describing the flow structures using the numerical time-dependent vorticity fields. The computational analysis of the flow around static twin-box decks was addressed, for instance, in [de Miranda et al. \(2015\)](#), where the performance of RANS and LES approaches was compared; and in [He et al. \(2021\)](#), where interference effects were analysed by means of a parametric study dependent on the gap distance using 3D LES simulations for parallel bridges at 0° angle of attack. In the last reference, the focus was put on the mean and fluctuating pressure distributions and the flow structures for different gap distances between the parallel girders. The first numerical study on the VIV response of twin-box decks was the one reported in [Laima et al. \(2013\)](#), where a hybrid approach (experimental and computational) in which the flow structures at VIV were studied by means of 3D LES forced oscillations was adopted, setting the amplitude of the forced oscillations equal to the amplitude of the VIV-induced oscillations in wind tunnel tests. Recent applications of numerical methods in the VIV analysis of multi-box decks are mainly constrained to the qualitative description of the flow features and supporting the analysis of experimental results. In [Wang et al. \(2022\)](#), 2D URANS models of a static triple-box girder considering the reference deck geometry, and another featuring grid plates for mitigating VIV, were reported for the visualisation of the flow structures and their qualitative description. A similar approach was adopted in [Chen et al. \(2022\)](#), where 2D URANS simulations were applied for obtaining the mean and fluctuating pressure coefficients along with the flow features for different deck geometries. A fluid–structure interaction (FSI) simulation was conducted for one deck geometry at only one reduced velocity, examining the vortical

structures and the amplitude of oscillation which was 10% lower than the experimental reference. Finally, another simulation was completed for the cross-section fitted with splitter plates, reporting the decrease in the amplitude of oscillation. [Zhang et al. \(2023\)](#), studied by means of CFD methods a triple-box deck, considering curved and linear cross-section geometries for the boxes. The targeted aerodynamic outputs were the pressure distributions, flow structures and the contribution of the aerodynamic work at VIV. For the FSI simulations, a 2D URANS approach was considered. The amplitudes of oscillation obtained for different reduced velocities were validated with experimental tests, showing qualitative agreement, although with substantial discrepancies in the amplitude value between CFD and experimental data for certain reduced velocities.

Previous CFD-based studies addressing the VIV response of multi-box bridge decks have been constrained to 2D URANS simulations, providing the qualitative description of flow features around the nominal configuration of the deck, and in some cases, for alternative arrangements where the deck was fitted with a certain type of aerodynamic countermeasures. Similarly, mean and fluctuating pressure distributions have been reported, and only exceptionally the energy distributions around the deck boxes were provided ([Zhang et al., 2023](#)). In most of the cases, the validation of the CFD simulations was based on a limited number of outputs considering only a qualitative approach, with shortcomings in accuracy for the few cases where the amplitudes of oscillation of twin-box decks have been obtained using CFD models.

This work contributes to current research efforts by soundly addressing spatial and temporal discretisation sensitivity in 3D LES simulations, and considering a triple validation based on comparison between numerical results and experimental data for mean force coefficients and Strouhal number, mean and fluctuating pressure distributions, and VIV amplitude vs. reduced velocity. Furthermore, this piece of research exploits CFD-generated data for the detailed analysis at VIV of the complex interplay between the flow and the twin-box deck as a whole, but also at the level of each individual box. In this manner, for instance, spectra can be analysed for time-series of the complete deck and individual boxes, hysteresis loops are obtained and work distributions around boxes can be studied for different reduced velocities, depicting clear trends in the VIV response depending on the behaviour of each individual box. Therefore, the paper explores the ability of 3D LES simulations for assessing the VIV features in this complex geometry. Furthermore, this CFD model exploits the use of synchronous data recording, including flow visualisation, that would be complex and costly at a wind tunnel facility. It is worth mentioning that the use of CFD techniques would allow the analysis of a larger number of possible solutions at the deck design stage, keeping the experimental campaigns for the most promising alternatives according to numerical models, and hence providing designs with better performance under VIV.

The remaining of the paper is organised as follows: Firstly, the formulation defining the governing equations for the LES modelling approach is presented along side with the verification method and other relevant formulae. Afterwards, the characteristics of the different meshes used in the verification task are presented along with the computational model adopted. Section 4, addresses the spatial and temporal verification, as well as validation, using well established metrics. In Section 5, the results of the simulations undergoing free heave oscillation are reviewed and discussed considering aspects such as force and pressure spectra, correlations, time-frequency analysis, work or the description of the main flow features. Finally, the main conclusions of this piece of research are summarised in Section 6.

2. Formulation

2.1. Governing equations

The Navier–Stokes equations, which enclose the fundamental fluid mechanics principles, can be modelled using different approaches. In

the frame of the present study, these have been modelled following the Large Eddy Simulation (LES) formulation. In this approach, the time-dependent Navier–Stokes equations are spatially filtered, resulting in the following pair of equations, as described in Bruno et al. (2010):

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (1)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial \bar{u}_i \bar{u}_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\nu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) + \tau_{ij}^s \right], \quad (2)$$

where $\bar{u}_i$ is the filtered velocity, $\bar{p}$ is the filtered pressure, x_i represents the space coordinates, t is the time, ν is the kinematic viscosity and ρ is the fluid density.

Using the Boussinesq assumption, the sub-grid stress tensor is expressed as:

$$\tau_{ij}^s = \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right), \quad (3)$$

where ν_t is the subgrid-scale turbulent viscosity.

The turbulence model selected for the present study is the Smagorinsky model (Smagorinsky, 1963). This model is based on the assumption of equilibrium between the small and resolved scales, dissipating the small ones the energy extracted from the resolved ones. Moreover, it obeys the following equation:

$$\nu_t = (C_S \Delta)^2 \left(2 \overline{S_{ij} S_{ij}} \right)^{1/2}, \quad (4)$$

where C_S is the Smagorinsky constant, $\overline{S_{ij}}$ is the filtered strain rate tensor and Δ is the characteristic spatial length of the filter, related to the mesh size, and defined as the cubic root of the mesh cell volume. The analysis of the impact caused by the selection of different LES turbulence models and the characteristic spatial lengths, can be found in Álvarez et al. (2021) for the same deck geometry under study in this work.

2.2. Force coefficients, Strouhal number and pressure coefficient

The time-dependent force coefficients (drag (C_d), lift (C_l) and moment (C_m)) along with the Strouhal (St) number, which can be referred also as integral parameters, are calculated according to:

$$C_d = \frac{F_D}{\frac{1}{2} \rho U^2 C}, \quad C_l = \frac{F_L}{\frac{1}{2} \rho U^2 C}, \quad C_m = \frac{M}{\frac{1}{2} \rho U^2 C^2}, \quad St = \frac{f D}{U}, \quad (5)$$

where C stands for the width of the single box (see Fig. 1), D is the depth of the single box, ρ is the air density, U is the free-stream velocity, f is the dominant frequency of the lift coefficient, and F_D , F_L and M are the time-dependent drag and lift forces and moment per unit of span length, which are calculated as the spanwise averaging of the integration of the pressure and viscous forces along the twin-box surfaces at each time-step. The sign convention for the force coefficients is presented in Fig. 1. The positive directions for the lift and moment are the upward direction and clockwise rotation respectively. The positive drag is taken in the along-wind direction, which is coincident with the positive x -axis in Fig. 1. The relationships between the different dimensions presented in Fig. 1 are: $B/C = 2.73$, $C/D = 5.40$ and $G/D = 3.96$.

In the following, the time averaged force coefficients values will be referred as $\bar{C}_i$ ($i = d, l, m$) and their standard deviations as $\tilde{C}_i$, which are calculated according to Eq. (6):

$$\tilde{C}_i = \sqrt{\frac{1}{n} \sum_{k=1}^n (C_{ik} - \bar{C}_i)^2}, \quad (6)$$

being C_{ik} the discrete force coefficient instantaneous values along the studied time interval.

The mean pressure coefficient ($\bar{C}_p$) and its standard deviation ($\tilde{C}_p$) are calculated as:

$$\bar{C}_p = \frac{\bar{p}}{\frac{1}{2} \rho U^2}, \quad (7)$$

$$\tilde{C}_p = \frac{\tilde{p}}{\frac{1}{2} \rho U^2}. \quad (8)$$

In the above equations, $\bar{p}$ is the time averaged pressure and $\tilde{p}$ is the standard deviation of the pressure at each point on the deck surface. The distributions of $\bar{C}_p$ and $\tilde{C}_p$ around the deck help to understand the resultant mean and fluctuating force coefficients, and the fundamental flow features in the vicinity of the boxes.

2.3. Spatial and temporal discretisations. Uncertainty assessment

The method described in Celik et al. (2008) for the calculation of the uncertainty associated to the level of spatial or temporal discretisations is based on power expansion series, whose basic equation for the estimation of the discretisation error (ϵ) is:

$$\epsilon_\phi \simeq \delta_{RE} = \phi_i - \phi_0 = \alpha h_i^p, \quad (9)$$

where ϕ_i is the aerodynamic output for which the uncertainty is calculated, ϕ_0 is the estimated exact solution, α is a constant, h_i is the representative cell size or time-step length and p is the observed order of grid convergence.

In order to apply Eq. (9), two assumptions must hold (Eça and Hoekstra, 2014): the finite volume grids must be in the asymptotic range to guarantee that the leading term of the power series expansion is sufficient to estimate the error, and the level of refinement of the mesh can be represented by a single parameter, that is a representative cell size.

The representative cell size is calculated as:

$$h = \left[\frac{1}{N} \sum_{i=1}^N (\Delta V_i) \right]^{1/3}, \quad (10)$$

where N is the overall number of mesh elements and ΔV is the cell volume.

The application of this method requires of three meshes with increasing refinement level, therefore $h_1 < h_2 < h_3$, and thus subindex 1 refers to the fine mesh and 3 to the coarse one. The grid refinement factor $r = h_3/h_1$, has to be greater than 1.3 (Celik et al., 2008).

The apparent order of the method (p) is calculated as shown in Eqs. (11)–(14):

$$p = \frac{1}{\ln(r_{21})} \left[\ln |e_{32}/e_{21}| + q(p) \right], \quad (11)$$

$$q(p) = \ln \left(\frac{r_{21}^p - s}{r_{32}^p - s} \right), \quad (12)$$

$$s = 1 \cdot \text{sign}(e_{32}/e_{21}), \quad (13)$$

$$r_{21} = h_2/h_1; \quad r_{32} = h_3/h_2; \quad e_{21} = \phi_2 - \phi_1; \quad e_{32} = \phi_3 - \phi_2. \quad (14)$$

Values of $e_{32}/e_{21} < 0$ are indicative of oscillatory convergence, being its sign the one defining the sign of s in Eq. (13). The method continues calculating the extrapolated values as:

$$\phi_{ext}^{21} = (r_{21}^p \phi_1 - \phi_2)/(r_{21}^p - 1); \quad \phi_{ext}^{32} = (r_{32}^p \phi_1 - \phi_2)/(r_{32}^p - 1). \quad (15)$$

This method, along with the apparent order p , also requires the approximate relative error (e_a , see Eq. (16)), the extrapolated relative error (e_{ext} , see Eq. (17)) and the grid convergence index for the fine mesh (GCI_{fine} , see Eq. (18)). These are calculated as follows:

$$e_a^{21} = \left| \frac{\phi_1 - \phi_2}{\phi_1} \right|, \quad (16)$$

$$e_{ext}^{21} = \left| \frac{\phi_{ext}^{12} - \phi_1}{\phi_{ext}^{12}} \right|, \quad (17)$$

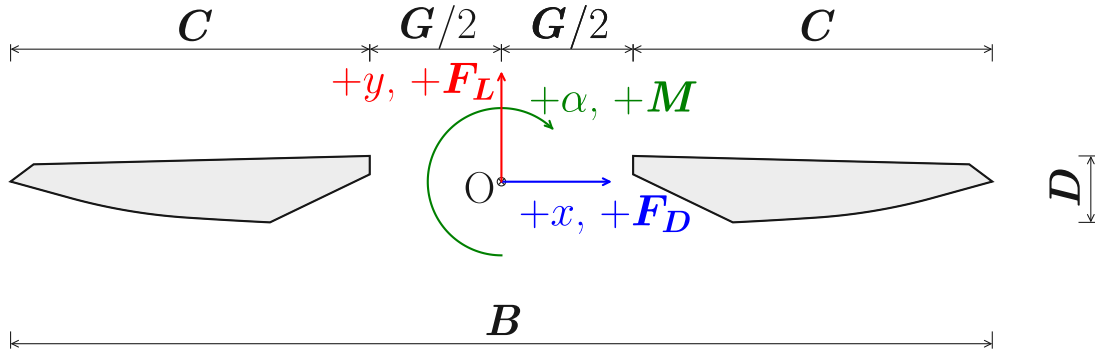


Fig. 1. Sign convention.

$$GCI_{fine}^{21} = \frac{1.25e_a^{21}}{r_{21}^p - 1}. \quad (18)$$

The numerical uncertainty for each grid is evaluated according to Eq. (19), which to some extent shows the reliability of the presented results, as for the case of experimental campaigns. For further information about this method, the interested reader is referred to Celik et al. (2008) and Roache (2009).

$$U_{fine}^{21} = \pm GCI_{fine}^{21} \phi_1. \quad (19)$$

2.4. Validation metrics

In order to assess the level of agreement between the reported numerical results and available experimental data different metrics have been proposed in the literature. The formulation of the metrics adopted herein are presented next:

- Hit Rate:

$$HR = \frac{1}{N_m} \sum_{n=1}^{N_m} \delta_{HR,n}, \quad \delta_{HR,n} = \begin{cases} 1, & \frac{|Sim_n - Exp_n|}{|Exp_n|} \leq \Delta_r \text{ or } |Sim_n - Exp_n| \leq \Delta_a, \\ 0, & \text{otherwise,} \end{cases} \quad (20)$$

where N_m is the number of measurements, Sim are the numerical values, Exp are the experimental values, Δ_r is the allowed relative difference and Δ_a is the allowed absolute difference.

- Correlation coefficient:

$$R = \frac{cov(Sim_n, Exp_n)}{\sigma(Sim_n) \sigma(Exp_n)}, \quad (21)$$

where cov refers to the covariance of the two variables and σ represents its standard deviation.

- Fraction of data within a factor α :

$$FAC_\alpha = \frac{1}{N_m} \sum_{n=1}^{N_m} \delta_{FAC,n}, \quad \delta_{FAC,n} = \begin{cases} 1, & \frac{1}{\alpha} \leq \frac{Sim_n}{Exp_n} \leq \alpha, \\ 0, & \text{otherwise,} \end{cases} \quad (22)$$

- Index of agreement:

$$IoA = 1 - \frac{\sum_{n=1}^{N_m} (Sim_n - Exp_n)^2}{\sum_{n=1}^{N_m} (|Sim_n - \overline{Exp_n}| + |\overline{Exp_n} - Exp_n|)^2}, \quad (23)$$

where $\overline{Exp_n}$ is the mean of the experimental values.

In all the metrics defined, the closer the value is to 1.0, the better the agreement between the experimental and the numerical data. For the index of agreement IoA , a value of 1.0 means a perfect match between data series.

2.5. Correlation of overall and local lift coefficient, contribution of local lift coefficient and non-dimensional work done by lift coefficient at VIV

The correlation coefficient is a statistical parameter that provides values between -1 to 1 , aiming at quantifying the strength of the interdependency between two variables. The value of zero means that the two variables are uncorrelated, and for the case of linear and Gaussian stochastic processes this also indicates independence. Moreover, the correlation coefficient provides information about the phase-lag between the two variables under study: when two variables with pure sinusoidal time-histories are considered, the correlation coefficient is equal to $\cos\phi$, being ϕ the phase-lag between the two time-histories. Correlation also has a physical meaning in fluid-structure interaction phenomena since, according to Ricciardelli (2010), the lower the correlation, “the greater the reduction of the fluctuating loads and dynamic response”. The correlation coefficient is calculated as:

$$R_i = \frac{cov(C_l, C_{l,i})}{\sigma(C_l) \sigma(C_{l,i})}. \quad (24)$$

where C_l is the time-history of the overall lift coefficient, $C_{l,i}$ is the time-history of the lift coefficient contribution of each probe (see Fig. 2).

The lift coefficient associated with each probe $C_{l,i}$ on the deck surface is calculated as indicated in Eq. (25):

$$C_{l,i} = -C_{p,i} \frac{S_i}{C^2} \langle \mathbf{y}, \mathbf{n} \rangle, \quad (25)$$

where $C_{p,i}$ is the pressure coefficient at probe i , S_i is the tributary area of probe i , $\mathbf{y}$ is the positive vertical axis (see Fig. 1), $\mathbf{n}$ is the outward surface normal vector and $\langle \cdot, \cdot \rangle$ represents the inner product.

The contribution of each probe to the resultant deck lift coefficient is calculated as indicated in Eq. (26), where the correlation coefficient of the overall and local lift coefficients is multiplied by the standard deviation of the local lift coefficient (Hu et al., 2022).

$$\hat{C}_i = R_i \sigma(C_{l,i}). \quad (26)$$

In this piece of research, the non-dimensional work generated by the lift coefficient or the lift coefficient associated with each probe while undergoing heave oscillations is calculated as:

$$W(t_j) = \frac{C_*(t_j) + C_*(t_{j+1})}{2} \frac{C}{D} \frac{Y(t_{j+1}) - Y(t_j)}{D}, \quad (27)$$

where C_* is the lift coefficient of each box or the overall deck, or the lift coefficient associated with each probe, $W(t_j)$ is the non-dimensional work of the lift coefficient at the current time step, and $t(j+1)$ refers to

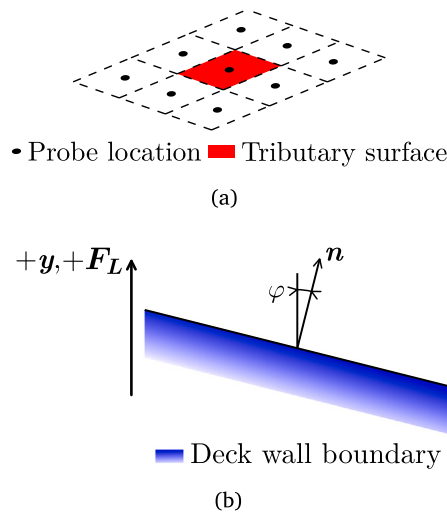


Fig. 2. (a) Sketch of pressure probes on a generic surface and tributary area considered for the calculation of local $C_{l,i}$; (b) $\mathbf{n}$ and $\mathbf{y}$ vectors convention.

Table 1

Overall fluid domain dimensions. B is the deck width, and C and D are the width and height of each individual box. (See Fig. 3 for the graphical definition of the flow domain variables.)

Ax	Ay	Dx	Dy	Dz
15C	15C	40C + B	30C + D	C

the values at the next time step. The cumulative non-dimensional work is:

$$W^*(t_j) = \sum_{i=1}^j W(t_i). \quad (28)$$

3. Flow domain and computational modelling

The twin-box deck under study corresponds to the VIV-prone bare cross-section of the Stonecutters bridge without considering the transversal beam joining both decks. In the following, the flow domain and the characteristics of the finite volume mesh are described, along with the computational modelling approach. These 3D LES simulations are solved with the CFD software OpenFOAM 5.0 (Greenshields, 2017).

3.1. Mesh characteristics

The overall fluid domain is depicted in Fig. 3(a), and its dimensions are reported in Table 1. The spanwise length considered in this study is equal to the width of a single box (C), following the criteria used in authors' previous works (Álvarez et al., 2019, 2021). The flow inlet is located at the left hand side boundary (Fig. 3(b))

The mesh topology used for the discretisation of the fluid domain is a structured quadrangular mesh for the boundary layer region around the boxes and the spanwise dimension, meanwhile for the rest of the XY plane an unstructured quadrangular mesh is adopted. As in previous works (Álvarez et al., 2021), the XY plane is subdivided in different regions aiming at efficiently handle the mesh density grading. The fundamental outline of the extruded 2D mesh in the XY plane is presented in Fig. 3(b).

In Table 2, the side lengths of the cells along the external perimeter of the different zones (see Fig. 3) for the three meshes considered in the verification study are presented. The sizes of the elements in these zones increase as they are located further away from the deck. It should be noted that the size provided for Zone L (the wake) is the one along its

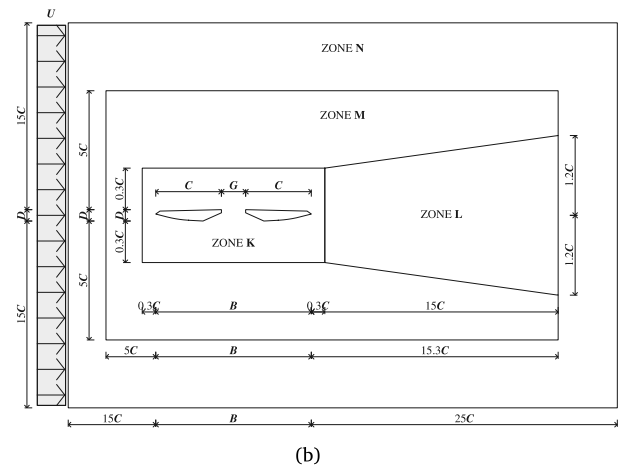
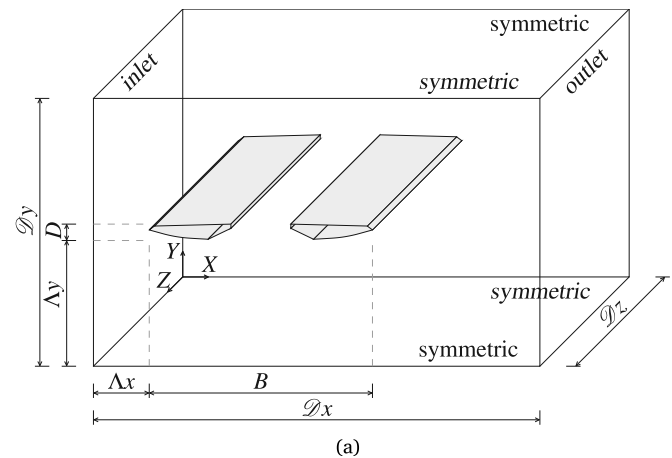


Fig. 3. (a) Overall fluid domain and (b) different zones in which the XY plane mesh is subdivided. (Not to scale).

Table 2

Cell sizes on the boundaries of the different mesh zones K, L, M and N. All the dimensions are non dimensional, after having been divided by C. (The numbers between parenthesis are the subindexes referring to the discretisation level in the verification study as per Section 4.1).

Mesh	Zone K	Zone L	Zone M	Zone N
Fine (1)	0.008	0.083	0.208	0.418
Medium (2)	0.021	0.145	0.330	0.800
Coarse (3)	0.031	0.200	0.450	1.000

right side vertical boundary, meanwhile the cell sizes along the upper and lower boundaries grow from left to right, starting with the size of the buffer zone (Zone **K**), following a geometric series until the cell upper side length reaches the size of Zone **L** vertical right boundary discretisation. Also, along the right limits of Zone **M**, the cells increase their sizes from the one assigned for the Zone **L** boundary, up to the one specified for Zone **M**. It should be noted that the 3D meshes are generated by extrusion.

The characteristics of the boundary layer mesh and the spanwise discretisation, which are the same for the three meshes considered in the verification study reported in Section 4.1, are presented in Table 3; meanwhile, the number of elements in the different zones are reported in Table 4 for one XY plane mesh layer. It may be noticed that the closer each zone is to the deck, the higher the number of elements is, aiming at more accurately simulate the flow phenomena that are taking place in the vicinity of the boxes and in their wakes. Graphical details of the medium mesh in the XY plane are shown in Fig. 4.

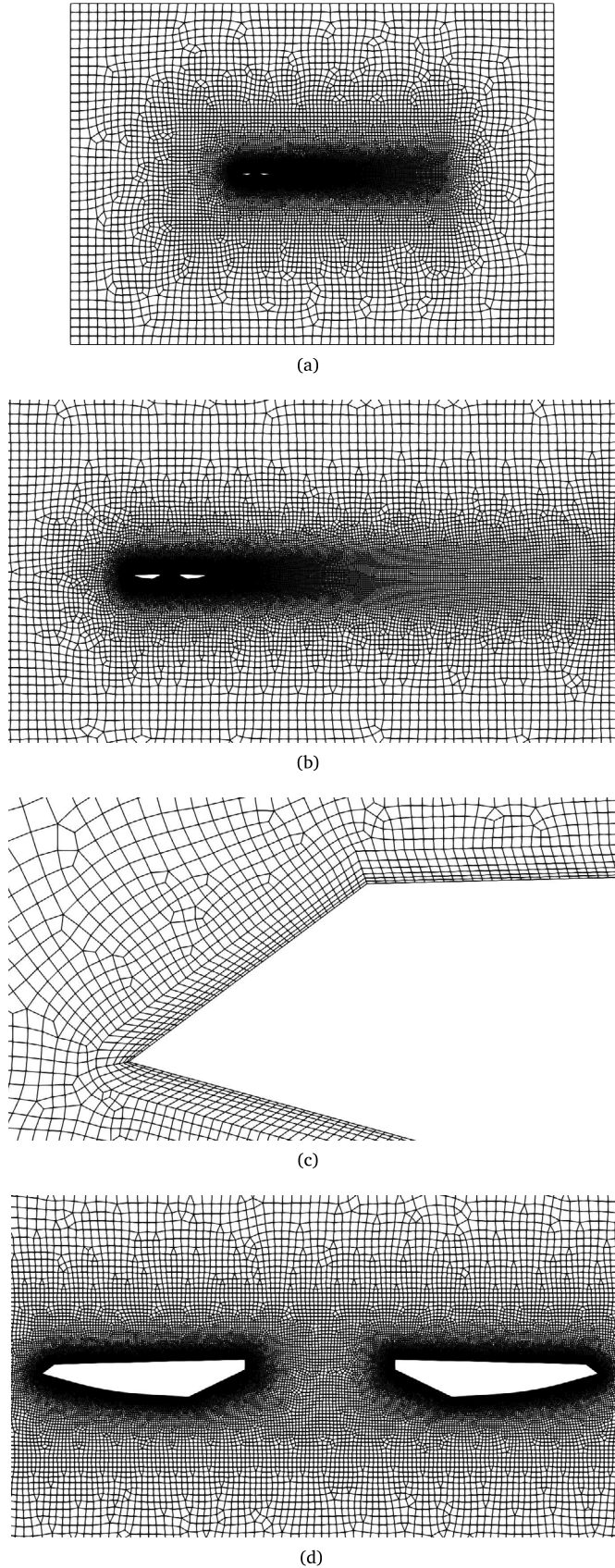


Fig. 4. Mesh discretisation details of the medium mesh. (a) overall fluid domain, (b) buffer zone and wake, (c) mesh around the deck and (d) boundary layer detail.

Table 3

Boundary layer mesh properties. The parameter y_1 is the height of the first row of elements of the boundary layer (BL) mesh, C is the width of a single box, x_1 is the length of the first row of elements in the BL, gr is the growth ratio of the elements in the BL, n_{BL} is the number of layers forming the BL mesh, y_{BL} is the total height of the BL mesh, δ_z is the length of the cell in the spanwise dimension and n_z is the number of elements in the spanwise dimension.

y_1/C	x_1/y_1	gr	n_{BL}	y_{BL}/C	δ_z/C	n_z
0.0007	4	1.32	6	0.0088	0.021	48

The y^+ values obtained for the three meshes at $Re_D = 4.48 \cdot 10^4$, for the static deck, at 0° angle of incidence, are presented in Table 5. For all the cases, in both boxes, the mean $\overline{y^+}$ value is very close to 1, and less than 4.5% of the elements show y^+ values higher than 4; hence no wall functions are adopted to model the boundary layer behaviour. These y^+ values were obtained as indicated in Bruno et al. (2010), although in this study the total height of the first row of the boundary layer mesh is adopted for their calculation.

Moreover, in Table 6, the time cost associated to the computation of one dimensionless time unit ($t^* = tU/C = 1$) for each mesh for the static case at 0° incidence is reported for the three meshes considered in the verification study. As expected, the higher the number of elements of the mesh, the higher the computational burden.

3.2. Computational modelling

The diffusive terms are discretised by using the second order central difference scheme, while the second order upwind differencing scheme is applied for the convective terms. Advancement in time is performed by the second order backward scheme. Finally, the pressure-velocity coupling is solved by the PIMPLE algorithm.

In the dynamic simulations, the Arbitrary Lagrangian Eulerian algorithm is applied to incorporate the movement of the mesh into the governing fluid equations. The deck is partially restrained, allowing displacements only in the heave degree of freedom, and its oscillation is modelled by a single-degree-of-freedom mass-spring-damper system. The coupling between the fluid, the structural system and the dynamic mesh is achieved by the adoption of the conventional serial staggered scheme.

The boundary conditions imposed in the simulations are Dirichlet type at the inlet for the velocity, meanwhile Neumann conditions are imposed for the inlet pressure. At the outlet, Dirichlet conditions are applied to the pressure, and Neumann type to the velocity. For the upper, lower and lateral faces, symmetric boundary conditions are adopted, as they provide physically sounder spanwise correlations (Álvarez et al., 2019). All the simulations consider smooth flow condition at the inlet ($I = 0\%$; I is the turbulence intensity). For the twin-box deck walls, no-slip and no-penetration boundary conditions are applied (Wilcox, 2012).

For all the simulations considered in this work, the convergence criteria for the mean and standard deviation of the force coefficients, as well as the Strouhal number, are based on the values of the residuals of the corresponding aerodynamic output. The residuals for a generic q response are calculated as $\varphi_{res} = |(q_n - q_{n-1})/q_n| \cdot 100$ (n is the number of sampling windows), and the simulations are extended until the residual value is lower than 5%, as suggested in Bruno et al. (2010). These residuals are obtained for increasing lengths of the sampling window T_n , with $T_0 = 50$ and $T_n = T_{n-1} + 50$ (in non-dimensional time units (tU/D))(see Fig. 5).

4. Verification and validation assessment

4.1. Spatial and temporal discretisation sensitivity studies

The selection of the mesh discretisation level is based on the level of uncertainty of the integral parameters for the twin-box deck under

Table 4

Number of elements of the different mesh zones in the XY plane. (n_z is the number of elements in the spanwise dimension and δ_z is the length of the cell in the spanwise dimension) (The numbers between parenthesis are the subindexes referring to the discretisation level in the verification study as per Section 4.1).

Mesh	Zone K	Zone L	Zone M	Zone N	Overall XY	n_z	δ_z/C	Overall
Fine (1)	102130	23415	42181	11403	179129	48	0.021	8598192
Medium (2)	43694	6275	13407	3668	67044	48	0.021	3218112
Coarse (3)	29247	2820	6592	2380	41046	48	0.021	1970208

Table 5

y^+ values for the set of meshes considered for the verification study in Section 4.1. (The numbers between parenthesis are the subindexes referring to the discretisation level in the verification study).

Mesh	Box	$\overline{y^+}$	$\% \overline{y^+} > 2$	$\% \overline{y^+} > 4$	$\% \overline{y^+} > 6$
Fine (1)	Upwind	0.996	7.65	4.00	1.69
	Downwind	1.146	13.06	2.71	1.57
Medium (2)	Upwind	0.938	7.31	3.50	1.41
	Downwind	1.244	13.21	3.05	1.48
Coarse (3)	Upwind	0.994	8.09	4.20	1.76
	Downwind	1.126	13.00	2.63	1.57

Table 6

Computational cost over a dimensionless time unit. (The numbers between parenthesis are the subindexes referring to the discretisation level in the verification study as per Section 4.1).

Mesh	# of elements	# of cores	Time per core [h]
Fine (1)	8598192	48	1.262
Medium (2)	3218112	48	0.423
Coarse (3)	1970208	48	0.254

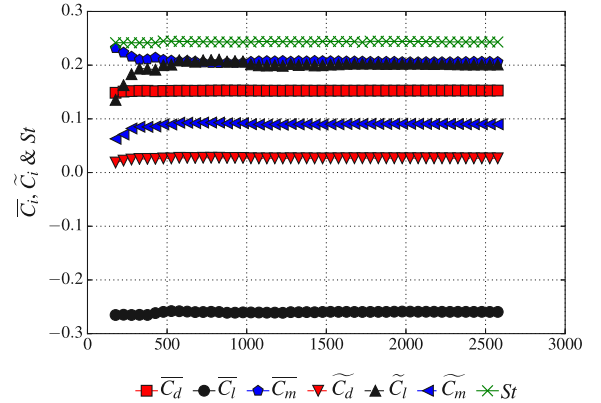
Table 7

Integral parameters in the spatial discretisation sensitivity study. (The numbers between parenthesis are the subindexes referring to the discretisation level in the verification study).

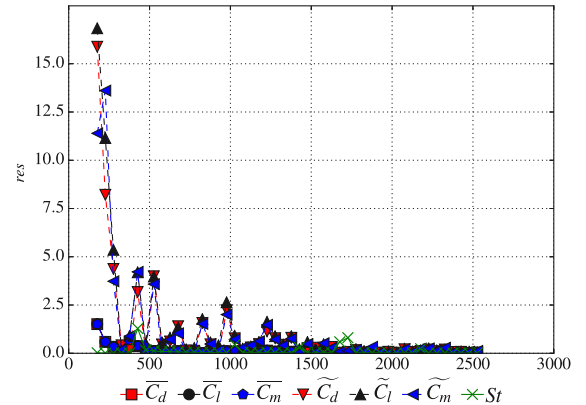
Mesh	Overall # elements	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	$\widetilde{C_d}$	$\widetilde{C_l}$	$\widetilde{C_m}$	St
Fine (1)	8598192	0.151	-0.263	0.226	0.023	0.166	0.076	0.247
Medium (2)	3218112	0.153	-0.260	0.206	0.027	0.202	0.090	0.244
Coarse (3)	1970208	0.154	-0.244	0.183	0.033	0.244	0.118	0.235

study. According to Roache (2009), the GCI method is of application to LES simulations as long as the spatial discretisation does not change dramatically for the coarse grid resolution case. It is to note that in this grid convergence study the boundary layer mesh and the spanwise discretisation is the same for the three levels of discretisation considered. The uncertainty calculation is done following the procedure proposed in Celik et al. (2008), and applied in Younis and Abrishamchi (2014), Sharma et al. (2020) and Sreejith et al. (2020), among others. The three meshes introduced in Section 3.1 are considered in the study, along with three different maximum Courant (Co) numbers to assess the time-step size impact in the uncertainty of the integral parameters. The Reynolds number adopted in the study is $Re_D = 4.48 \cdot 10^4$ and the angle of attack is 0° . In Table 7, the integral parameters for the three meshes with different spatial discretisation levels ($Co = 1.0$), alongside with their number of elements, are presented.

In Table 8, the uncertainty values obtained applying the procedure proposed by Celik et al. (2008) for the three levels of spatial discretisation considered herein are presented (further detail about the calculations can be found in Appendix A). For the medium mesh, the higher uncertainty is obtained for the standard deviation of the lift coefficient: $U_{medium}^{32}(\overline{C_l}) = 6.576 \cdot 10^{-02}$. The comparison of the values in Table 7 for the medium and fine mesh show a change in the fluctuating lift from 0.20 to 0.17, which does not represent a substantial change in the aerodynamic response. Previous work by the authors (Álvarez et al., 2019) showed even larger decreases in the fluctuating lift coefficient than in the current case for higher mesh refinement levels, which were linked with a minor decrease in fluctuating pressure coefficients.



(a)



(b)

Fig. 5. (a) Integral parameters evolution and (b) Integral parameters residuals for increasingly longer sampling windows, for the medium mesh and maximum Courant number $Co = 1$.

Based on the uncertainty study, the selected mesh is the medium one, as it presents a lower computational burden with respect the fine mesh (see Table 6), and provides low uncertainty values (U_{medium}^{32} in Table 8). Once the mesh is selected, the temporal discretisation sensitivity study is carried out for maximum Courant (Co) numbers 0.5, 1.0 and 2.5 (short, regular and large time discretisations). The values of the integral parameters obtained for each Courant number are presented in Table 9. Applying the procedure proposed in Celik et al. (2008) to the temporal discretisation, the uncertainty values for each integral parameter are presented in Table 10 (the complete calculations can be found in Appendix A). The uncertainty values associated to the temporal discretisation obtained in this study are low for all the maximum Courant numbers, specially for Co 0.5 and 1.0. As $Co = 1.0$ presents very low uncertainty values (see $U_{regular}^{32}$ in Table 10), and

Table 8Spatial discretisation uncertainty ($Co = 1.0$) (See Table A.1 for the complete calculations).

	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	St
U_{fine}^{21}	1.292×10^{-03}	7.701×10^{-05}	1.180×10^{-02}	1.417×10^{-03}	2.030×10^{-02}	2.658×10^{-03}	3.398×10^{-04}
U_{medium}^{32}	3.441×10^{-03}	3.512×10^{-03}	3.722×10^{-02}	6.506×10^{-03}	6.576×10^{-02}	2.019×10^{-02}	4.459×10^{-03}
U_{coarse}	1.511×10^{-02}	1.012	1.849×10^{-01}	7.819×10^{-02}	4.627×10^{-01}	5.546×10^{-01}	2.034×10^{-01}

Table 9Temporal discretisation sensitivity study for the integral parameters. (Co is the maximum Courant number) (The numbers between parenthesis are the subindexes referring to the discretisation level in the verification study).

	Co	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	St
Short (1)	0.5	0.153	-0.260	0.202	0.028	0.221	0.095	0.246
Regular (2)	1.0	0.153	-0.260	0.206	0.027	0.202	0.090	0.244
Large (3)	2.5	0.153	-0.265	0.203	0.026	0.197	0.088	0.241

demands a lower computational time burden, it is retained as the maximum Co for all the simulations reported afterwards.

4.2. Computational model validation with experimental data

In Table 11, the integral parameters obtained by adopting the selected mesh and time step in Section 4.1 (medium mesh and $Co = 1$) are presented and compared with the experimental values provided in Kwok et al. (2012). It can be noticed that the CFD results for the time-averaged force coefficients are close to the wind tunnel data, especially for the drag and lift coefficients. The difference in the mean moment coefficient is related with minor differences in the mean pressure coefficient distributions around each individual box that are amplified when the distance between the centre of pressure of each box and the centroid of the deck cross-section (see Fig. 1) are considered in the calculation of the resultant moment. Furthermore, it should be noted that the wind tunnel tests adopted for validation correspond to a sectional model with span length of $12.31C$, including the transversal beams connecting the two boxes, which are not considered in the CFD model.

Aiming at providing and objective assessment about the ability of the 3D LES model to accurately simulate the aerodynamic performance of the twin-box deck, the different metrics introduced in Section 2.4 are applied to the set of force coefficients and Strouhal number in Table 11. The results obtained in the statistical performance study are reported in Table 12. All the validation metrics present values close to 1.0, representing a good agreement between the experimental and numerical data. For the calculation of HR , the allowed relative difference is $\Delta_r = 0.25$, the same value adopted in Franke and Pham (2018) and Pham and Franke (2018), while the value $\alpha = 1.3$ has been selected as in Ricci et al. (2018).

After comparing the experimental and numerical results in Table 11 and obtaining the different values of the validation metrics, it can be concluded that the obtained integral parameters present a reasonable agreement with the available experimental data.

In Fig. 6, the mean and fluctuating pressure coefficient distributions for the selected mesh and maximum Courant number (medium mesh and $Co = 1.0$), alongside with the experimental data in Kwok et al. (2012) are depicted for comparison. For the $\overline{C_p}$ distribution, both numerical and experimental distributions present a good general agreement. The largest discrepancies in Fig. 6 can be appreciated over the windward upper panel of the upwind box, presenting the numerical distribution a suction plateau, also present in references such as Laima et al. (2018) for a twin-box deck, and Šarkić et al. (2015) for a mono-box deck, conducted at Re_D with the same order of magnitude as the one in the present study. For the distributions of the fluctuating pressure coefficient $\overline{C_p}$, the agreement around the upwind box is very good, while for the downwind box the numerical simulations provide slightly higher standard deviations in the vicinity of the gap between

boxes. This is arguably due to the absence of transversal beams in the CFD model, which lightly enhances the pressure fluctuations (Álvarez et al., 2021).

When the set of data under validation is composed by small magnitude values, the usual validation metrics deteriorate, being advised the graphical representation of the data in combination with sound engineering judgement for the assessment of the accuracy of the CFD simulations (Franke and Pham, 2017). This is the situation encountered for the mean and fluctuating pressure coefficients reported in Fig. 6; hence the validation metric proposed in this case is restricted only to the calculation of the correlation between experimental and CFD values. The results, close to 0.9 (0.904 for the $\overline{C_p}$ and 0.878 for the $\overline{C_p}$), are indicative of an accurate modelling of the time-dependent pressure distributions, and consistent with the good agreement obtained for the integral parameters discussed above.

5. CFD-based VIV analysis

In this section, the results of the CFD simulations under free oscillation in the heave degree of freedom for the Stonecutters bridge deck are presented. The mesh and maximum Co number adopted are the ones selected based on the sensitivity studies in Section 4.1 (medium mesh and $Co = 1.0$). The experimental campaign reported in Larsen et al. (2008) for the Stonecutters bare deck case has been selected for validation. The simulations are conducted adopting a CFD model reproducing the same geometric scale as in the experiments, which is 1:80. The Scruton number of both, the experiments and simulations, is $Sc = 32$ ($Sc = (4\pi M\xi)/(\rho L D^2)$, being M the mass of the system, ξ the relative damping, ρ the air density and L the spanwise dimension) (A. Larsen, personal communication, July 3–4, 2018). Furthermore, the swept Reynolds numbers for the VIV simulations are in the interval between $Re_D = 2.85 \cdot 10^3$ and $Re_D = 3.81 \cdot 10^3$. The frequency of the system in still air is $f_0 = 4.77$ Hz.

5.1. Amplitude of oscillation vs. reduced wind speed

The amplitudes of oscillation for the simulations under free oscillation in heave for several reduced velocities are presented and compared with the experimental data in Larsen et al. (2008) in Fig. 7, where the reduced velocity at vortex resonance is provided as reference. From this point onwards Y , refers to heave oscillation time-histories. The numerical results are able to reproduce the experimental data to a good extend. The simulations are capturing both the peak amplitude and the reduced wind speed range prone to VIV. It is noted that the amplitude of the peak oscillation is relatively low, about the 2% of the deck depth (see Fig. 8c). In the following subsections, the VIV characteristics are studied, analysing the main parameters, such as lift coefficient and heave time-histories. The main flow features along one cycle of VIV at peak amplitudes response ($U/(f_0 B) = 0.35$) are described based on the instantaneous vorticity snapshots over one cycle of oscillation, which are reported in Appendix B and Figs. B.1 and B.2.

5.2. Heave oscillation and lift coefficient time-histories, spectra and time frequency analysis

The heave oscillation and lift coefficient time-histories for all the studied reduced velocities $U/(f_0 B) = 0.30, 0.325, 0.35, 0.375$ and 0.4 are presented in Fig. 8. The reduced velocity $U/(f_0 B) = 0.35$ corresponds to the peak amplitude, meanwhile the others correspond to

Table 10
Temporal discretisation uncertainty (medium mesh). (See Table A.2 for the complete calculations).

	$\bar{C}_d$	$\bar{C}_l$	$\bar{C}_m$	$\bar{C}_d$	$\bar{C}_l$	$\bar{C}_m$	St
U_{short}^{21}	3.558×10^{-05}	1.616×10^{-05}	7.779×10^{-03}	2.946×10^{-04}	3.865×10^{-03}	1.587×10^{-03}	6.609×10^{-02}
$U_{regular}^{32}$	1.431×10^{-04}	2.174×10^{-04}	3.674×10^{-03}	3.538×10^{-05}	5.363×10^{-04}	3.180×10^{-04}	6.834×10^{-02}
U_{large}	3.636×10^{-03}	9.266×10^{-02}	1.119×10^{-02}	4.774×10^{-03}	5.121×10^{-02}	1.300×10^{-02}	7.311×10^{-02}

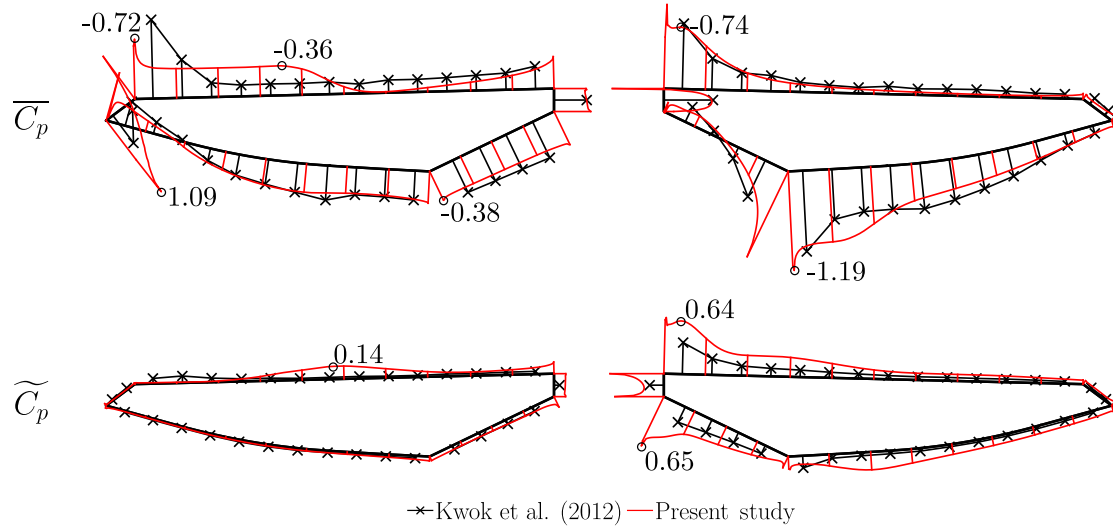


Fig. 6. Spanwise averaged $\bar{C}_p$ and $\bar{C}_l$ distributions. Gap width not to scale. (Negative values point outwards from the cross-section, and a value of unity for the coefficient is equal to the height of the box. The values in the plot correspond to CFD results marked with a black circle over the CFD results. The flow comes from the left hand side.)

Table 11
Integral parameters.

	$\bar{C}_d$	$\bar{C}_l$	$\bar{C}_m$	$\bar{C}_d$	$\bar{C}_l$	$\bar{C}_m$	St
Present study	0.153	-0.260	0.206	0.027	0.202	0.090	0.244
Kwok et al. (2012)	0.140	-0.234	0.268	-	-	-	0.278

Table 12
Metrics applied to the integral parameters.

Metric	Integral parameters
HR	1.000
R	0.992
FAC _{1.3}	0.75
IoA	0.974

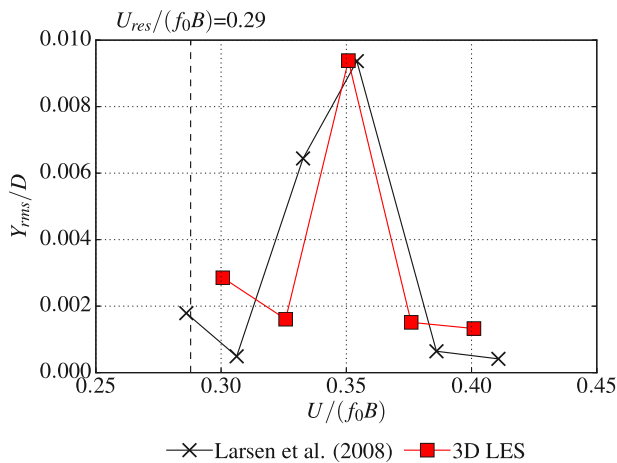


Fig. 7. Amplitudes of oscillation in heave at different reduced velocities.

reduced velocities at which the oscillation amplitudes are remarkably lower (see Fig. 7). In the following, the time histories of the overall deck and individual boxes are analysed aiming at identifying key features related with the VIV oscillation amplitude. This is further studied in the frequency domain and by time-frequency analyses.

The CFD approach selected herein enables the evaluation of the instantaneous lift coefficient acting on each individual box, as well as on the overall deck, which is equivalent to the synchronous measurements in wind tunnel campaigns (see for instance Hu et al., 2018). It is apparent that these are not pure sinusoidal signals, as they present multiple frequency peaks in their spectra (Fig. 9). The modulation present in the time-histories decreases with higher amplitudes of oscillation, but it is still present at the peak amplitude ($U/(f_0 B) = 0.35$). This phenomenon was also described in Mannini et al. (2016), where it is stated that the oscillations closer to the vortex shedding frequency present higher modulations, as for $U/(f_0 B) = 0.30$; but they are not so apparent as the reduced wind speed increases. It can be also noticed that the overall lift coefficient is very similar to the lift force acting on the downwind box, and that the phase-shift among signals does not remain constant through time.

In the frequency domain, the spectra presented in Fig. 9 permit the analysis of the dominant frequencies in the lift force coefficients of the complete deck and the individual boxes, and in the heave oscillation time-series. In this manner, the study of the energy content at the vortex shedding and natural frequencies for the set of studied reduced velocities allows a better understanding of the different levels of heave excitation and the identification of the lock-in condition.

In all the cases two strong peaks are identified: the peak associated with the smallest reduced wind velocity is related with the natural frequency of the system, meanwhile the peak associated with the higher reduced wind velocity is in correspondence with the vortex shedding frequency. In fact, as the reduced velocity increases (higher flow velocity) the non-dimensional frequency of the first peak in the spectrum decreases, as the natural frequency of the system remains constant, which progressively increases the separation between the peak frequencies associated with the heave oscillation and the vortex

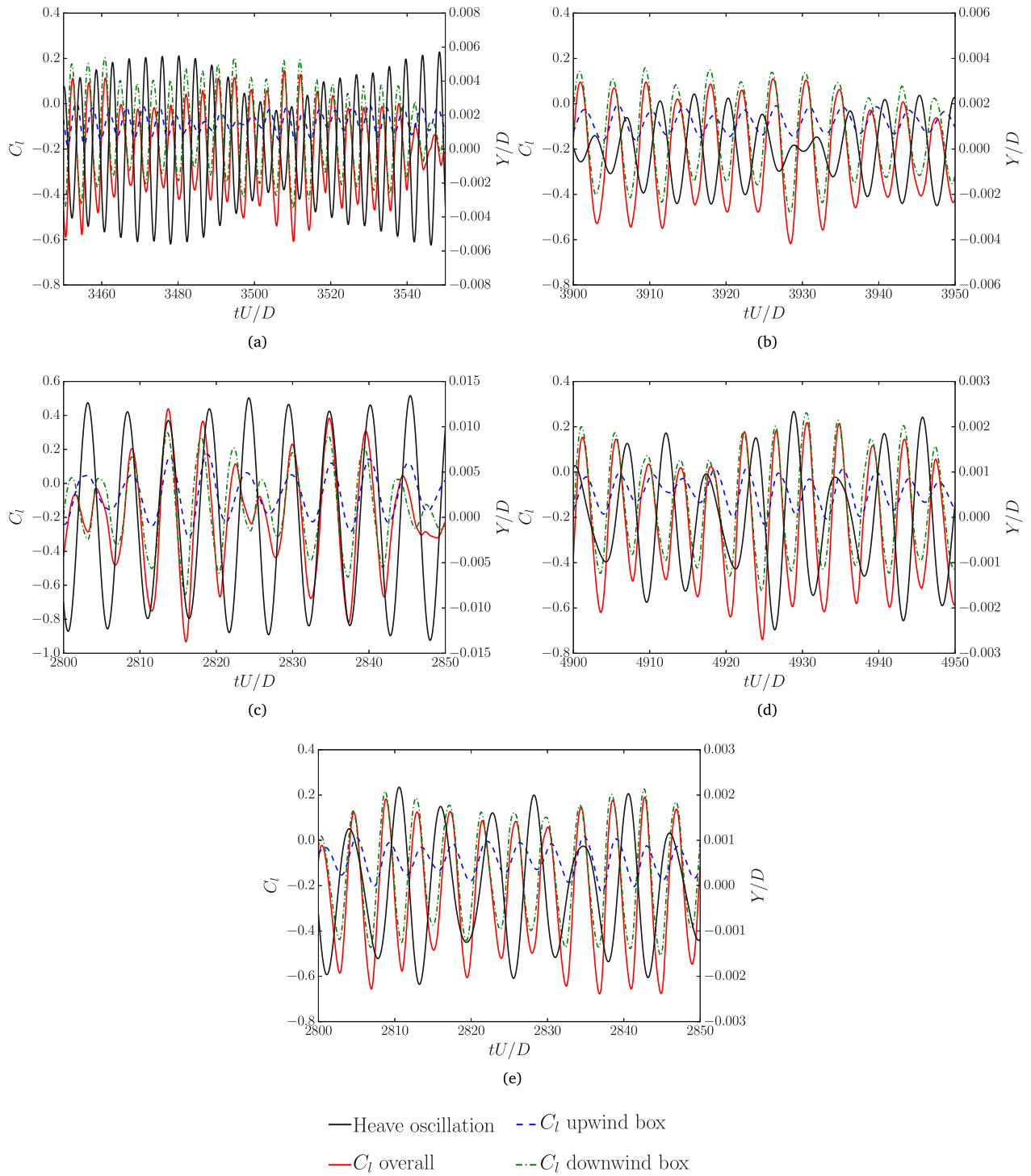


Fig. 8. Time-histories of the heave oscillation and lift coefficient a) $U/(f_0 B) = 0.30$, b) $U/(f_0 B) = 0.325$, c) $U/(f_0 B) = 0.35$, d) $U/(f_0 B) = 0.375$ and e) $U/(f_0 B) = 0.4$.

shedding. For the first studied reduced velocity ($U/(f_0 B) = 0.30$), Fig. 9a shows two very close distinctive peaks at the natural and vortex-shedding frequencies of the system in the lift coefficients and heave oscillation spectra. Focusing on the heave frequency response, the dominant frequency is the one associated with vortex shedding, indicative of this reduced velocity not belonging to the lock-in region of the system, but experiencing resonance with vortex-shedding. Figs. 9b, 9d and 9e show relatively weak peaks in the heave oscillation spectra at the natural and vortex shedding frequencies, consistent with minor flow induced oscillations in heave due to lift forcing featuring higher

energy at the vortex shedding frequency. However, the peak response case ($U/(f_0 B) = 0.35$) presents a strong peak in the heave spectrum at the natural frequency of the system, indicating VIV excitation (Fig. 9c left). It is interesting to analyse the response at each individual box in Fig. 9c right. In the upwind box, most of the lift force energy is concentrated at the natural frequency of the system as if the upwind box was experiencing classical lock-in; however, in the downwind box the energy content in the lift coefficient is similar at both the natural and the vortex shedding frequencies, and this is also the case for the overall deck. Therefore, the flow-induced deck response at this reduced

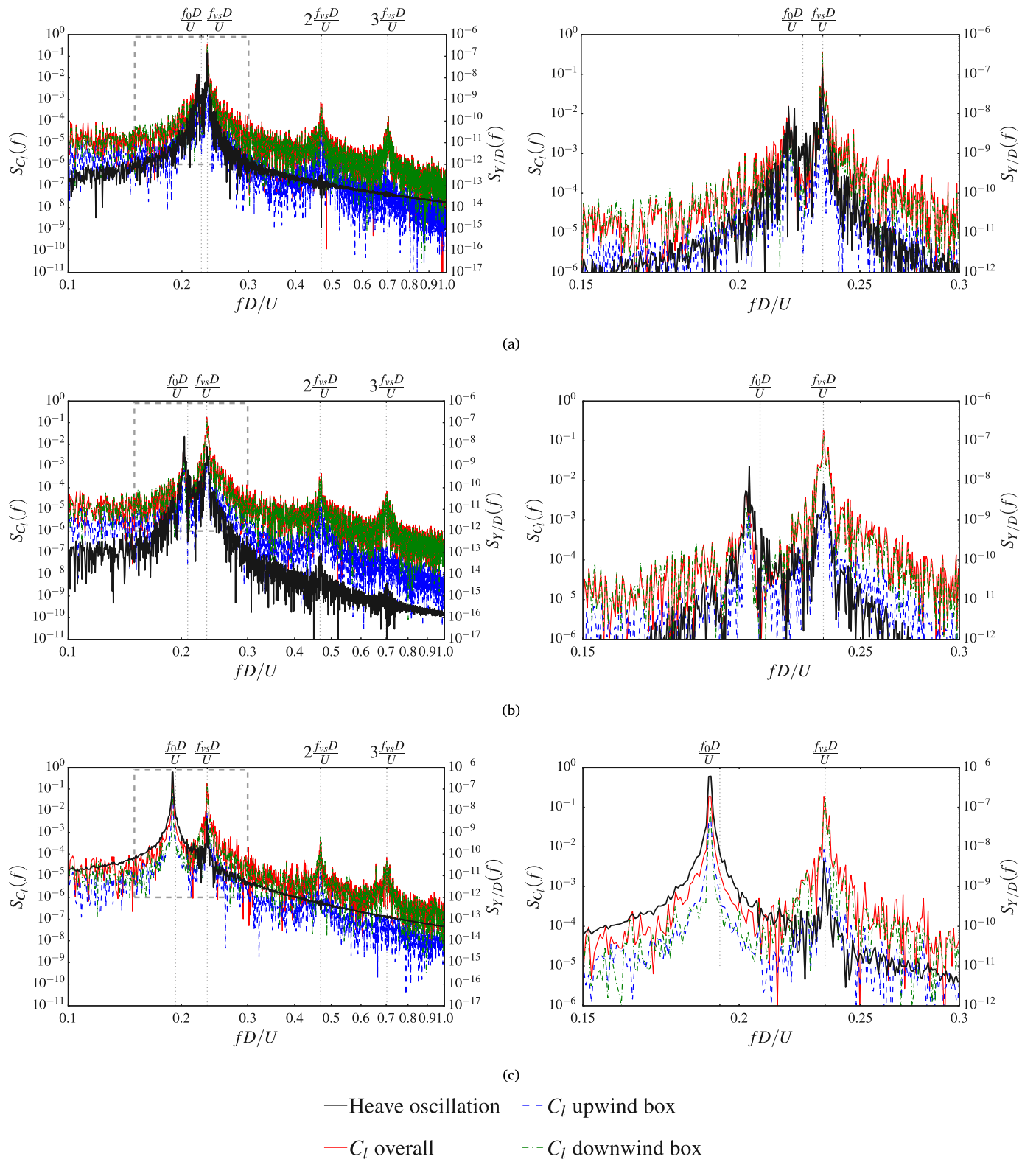


Fig. 9. Spectral density of the heave oscillation and lift coefficient (a) $U/(f_0 B) = 0.30$, (b) $U/(f_0 B) = 0.325$, (c) $U/(f_0 B) = 0.35$, (d) $U/(f_0 B) = 0.375$ and (e) $U/(f_0 B) = 0.4$. Note the different scales and ranges in the chart axes.

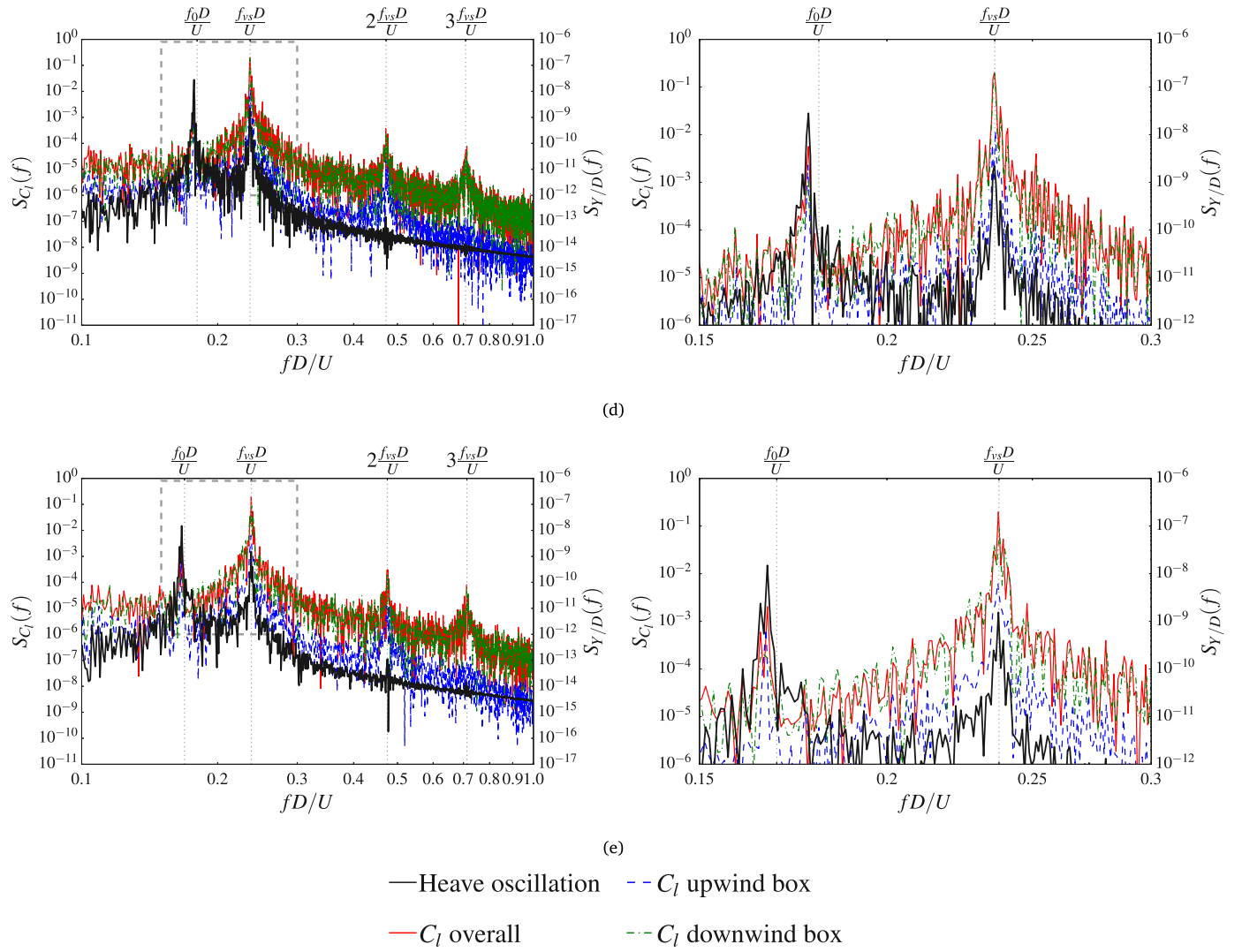


Fig. 9. (continued).

Table 13

Phase-lag between the heave oscillation and lift coefficient of the overall deck and for each box, within the whole range of analysed reduced velocities. (f_{peak} Y is the peak frequency from the heave oscillation, f_{peak} C_l is the peak frequency from the lift coefficient, γ_{Y-C_l} peak Y is the phase-lag between the heave oscillation and the lift coefficient at the Y peak frequency, γ_{Y-C_l} peak C_l is the phase-lag between the heave oscillation and the lift coefficient at the C_l peak frequency, f_0 is the natural frequency of the system).

	$U/(f_0 B)$	$f_{peak}(Y)/f_0$	$f_{peak}(C_l)/f_0$	γ_{Y-C_l} peak Y [°]	γ_{Y-C_l} peak C_l [°]
Overall	0.30	1.035	1.035	180.77	180.77
	0.325	0.980	1.123	3.99	179.57
	0.35	0.985	0.985	6.17	6.17
	0.375	0.984	1.310	4.56	180.07
	0.40	0.985	1.404	186.17	180.41
Upwind box	0.30	1.035	1.035	96.65	96.65
	0.325	0.980	1.123	-9.96	126.54
	0.35	0.985	0.985	-6.73	-6.73
	0.375	0.984	1.310	-4.60	137.26
	0.40	0.985	1.404	-9.71	139.88
Downwind box	0.30	1.035	1.035	193.09	193.09
	0.325	0.980	1.123	14.02	190.62
	0.35	0.985	1.210	17.94	196.50
	0.375	0.984	1.310	14.97	190.26
	0.40	0.985	1.404	20.63	189.63

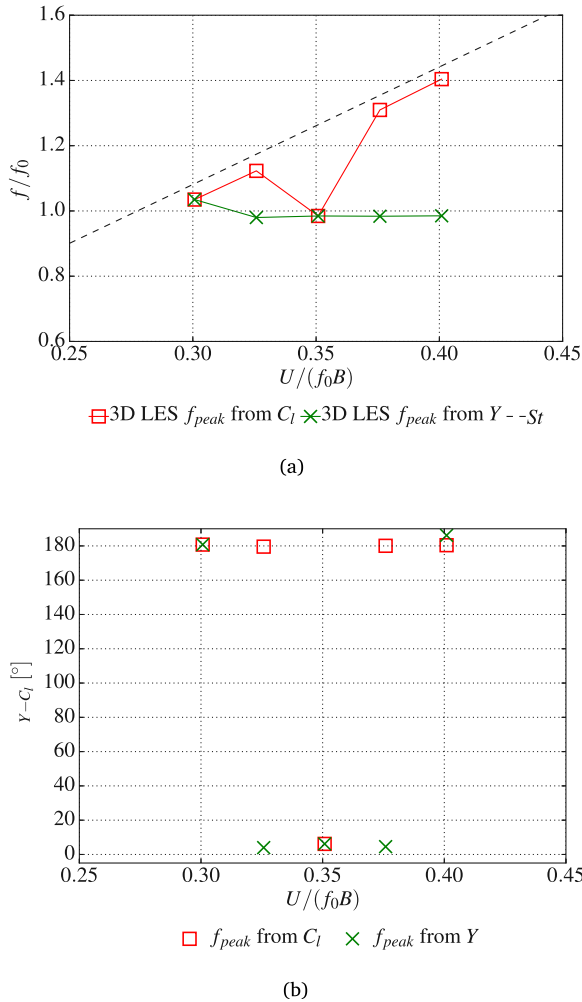


Fig. 10. VIV results. (a) peak frequencies from the Fourier analysis of the lift coefficient (f_{peak} from C_l) and the heave oscillation (f_{peak} from C_l) for each reduced velocity and their corresponding (b) phase-lag between the heave displacement and lift coefficient.

velocity is labelled as partial lock-in, since the classical behaviour described for the exciting force as perfectly locked at the natural frequency of the oscillator is not taking place. This is mainly due to the signature of the leeward box, which is being forced by the impinging vortices shed from the windward box, that have a non-negligible energy content at the vortex shedding frequency. This “decoupling” in the VIV response between the windward and leeward boxes is a consequence of the slot between the individual girders that strongly impacts the aeroelastic performance of twin-box decks. In all the studied cases the overall lift coefficient spectrum mainly resembles the one of the lift coefficient from the downwind box (see Fig. 9), which suggest that the oscillations may be controlled by the downwind box, as it is the one with higher energy content. Finally, it is noted that in all the studied reduced velocities, two distinctive super-harmonics at two and three times the vortex shedding frequency can be identified in the overall and downwind box lift force coefficients.

In Fig. 10(a), the principal frequency obtained from the Fourier analysis of the lift coefficient for the complete deck and the heave oscillation time-histories, are presented for the whole range of studied reduced velocities alongside with the Strouhal number (St) in Table 7 as reference. It can be noticed how the frequencies of the global

lift coefficient follow the trend of the St as the reduced velocity is increased, except for $U/(f_0 B) = 0.35$, for which the lift force frequency is the same as the natural frequency of the system. It is also observed in Fig. 10(a) that the heave peak frequencies for the lock-in region are slightly below the natural frequency of the system. This frequency of oscillation, slightly lower than the one in still air, has also been observed, for instance, in Hu et al. (2018), Bai et al. (2021) and Mannini et al. (2016), who assigns it to the low mass ratio of the system ($m^* = \rho V_B / (2M)$), where ρ is the air density, V_B is the overall volume of the decks and M is the overall mass of the system).

According to Nguyen et al. (2018) the limits of the lock-in region are attained when the lift and heave oscillation are out of phase. Therefore, in Fig. 10(b) the phase lags between the lift force and the heave oscillation are reported for the peak frequency in the lift coefficient time-series (red squares) and for the peak frequency in the heave displacement time-series (green crosses). Based on the phase-lags evaluated at the lift peak frequency, all the reduced velocities show out of phase lift forces against heave oscillation except for $U/(f_0 B) = 0.35$, for which partial lock-in was identified. On the other hand, when considering the peak frequency in the heave time histories (the natural frequency of the system), lift and heave displacements are nearly in phase for a wider range of reduced velocities (0.325–0.375), although a non-negligible heave oscillation is only obtained at $U/(f_0 B) = 0.35$, when the dominant peak frequency in the lift time-series is locked-in with the dominant frequency in the heave time-series. In Table 13, the values reported in Fig. 10, as well as the peak frequencies in the heave and lift coefficient time-histories of the individual boxes, and their corresponding phase-lags, are presented for further description of the VIV response.

The time-dependent lift coefficients and heave displacements presented in Fig. 8 showed oscillatory patterns that evolved over time, while the lift and heave oscillation spectra in Fig. 9 depicted two main frequency peaks at the system’s natural frequency and vortex shedding frequency, including also two additional super-harmonics of the vortex shedding frequency. Aiming at identifying the potential intermittence over time in the main frequencies identified in the spectra of Fig. 9, the Morlet wavelets for the overall lift coefficient and heave oscillation were obtained for the considered reduced velocities.

For all the studied reduced velocities above $U/(f_0 B) = 0.30$, the two distinctive peaks associated with the natural frequency of the system and vortex shedding frequency identified in Fig. 9 for the overall lift coefficient were also present, diminishing the intensity of the one corresponding with the natural frequency of the system as the amplitude of oscillation was reduced. Similarly, for the heave oscillation, two distinctive peaks corresponding with the natural frequency of the system and the vortex shedding frequency were also present, as previously observed in Fig. 9. No significant intermittent behaviour was identified for any of the studied cases.

5.3. Pressure spectra

In Fig. 11, the pressure spectra at three different positions on the deck (P01, P02 and P03) at reduced velocities 0.3 and 0.35 are reported to further outline the very different flow-induced responses obtained so far. P03 is located at the leeward end of the deck, far away from the gap-related vortex interaction, and downwind the separation point on the curved lower surface of the downwind box. No distinctive peaks can be identified in the spectra for both $U/(f_0 B) = 0.30$ and 0.35 (Figs. 11e and 11f), which means that the flow features governing the flow induced response are taking place upwind, notably in the gap between boxes instead of the eventual vortex shedding from the downwind box. Positions P01 and P02 show similar patterns depending on the reduced velocity. For $U/(f_0 B) = 0.30$ (Figs. 11a and 11c), featuring low oscillation amplitude, strong peaks at the vortex shedding frequency can be identified along with two other super-harmonics at P01 and P02. No significant peaks at the natural frequency of the system can

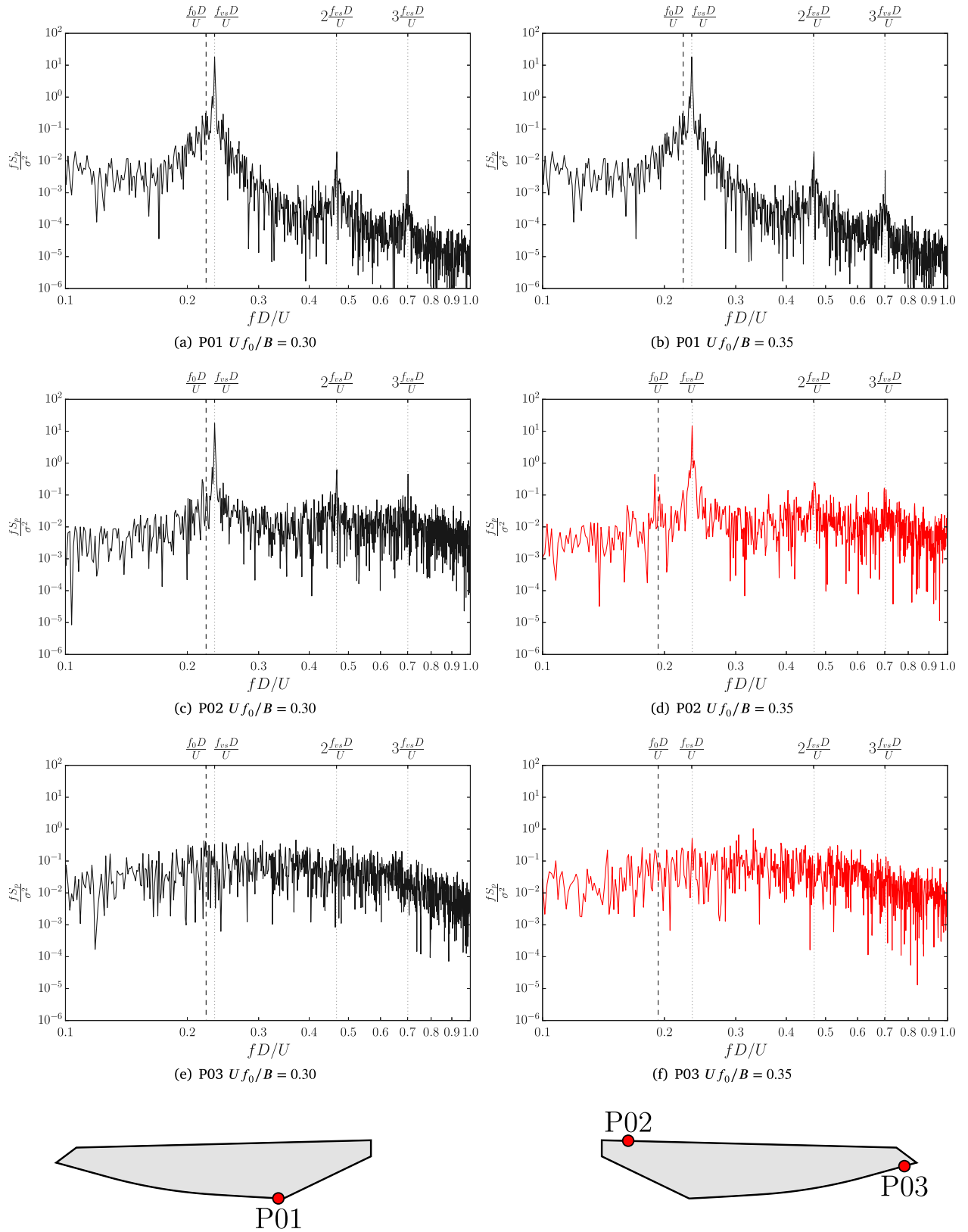


Fig. 11. Pressures spectra at three different positions along the overall deck. The flow comes from the left-hand side.

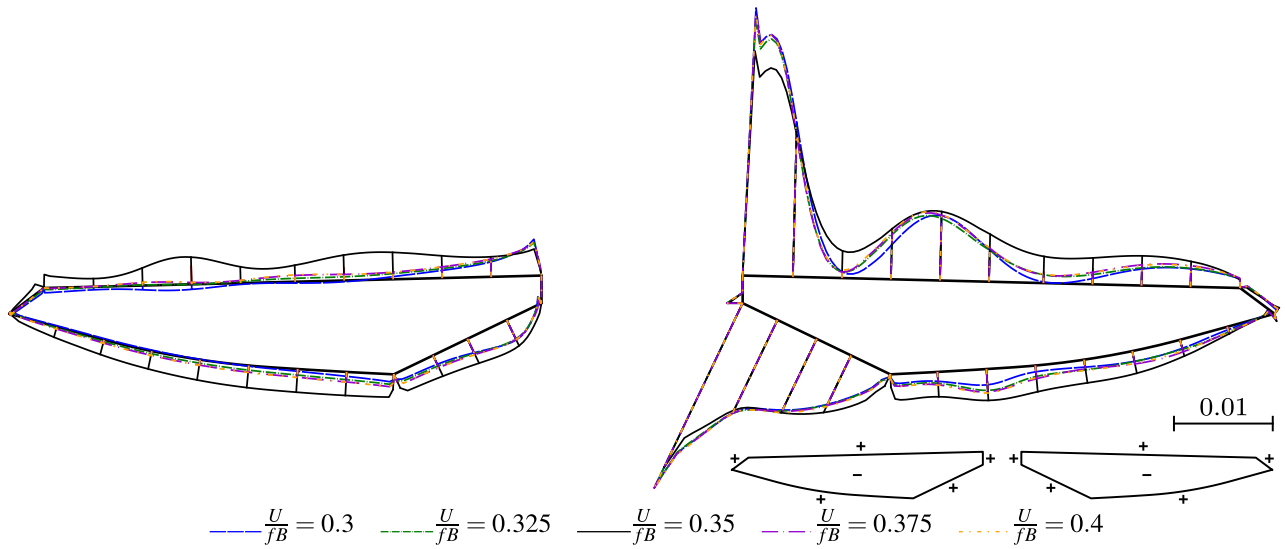


Fig. 12. Spanwise averaged contribution of the total lift coefficient and the local lift coefficient.

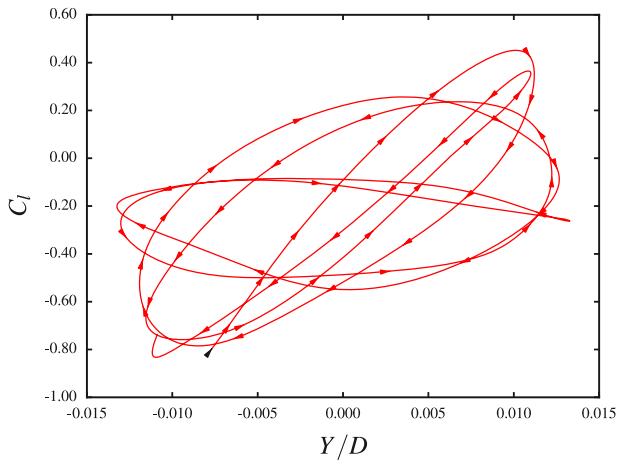


Fig. 13. Overall lift vs. Y/D trajectory over a few heave oscillation cycles for $U/(f_0B) = 0.35$.

be identified in that case. Consequently the pressure spectra at these positions, close to the gap, reflect the process of vortex shedding from the upwind box and the impingement on the downwind box. Conversely, for $U/(f_0B) = 0.35$ (Figs. 11b and 11d) the same P01 and P02 positions show different patterns in the spectra. In this last case, clear peaks at the natural frequency of the system are present, consistent with the VIV excitation. Furthermore, strong peaks at the vortex shedding frequency are also present, although the super-harmonics of the vortex shedding frequency are barely identifiable. It must be stated that for $U/(f_0B) = 0.35$, the peaks corresponding with the natural frequency of the system at P01 and P02 present similar amplitude for the absolute spectrum, meanwhile the peak related with vortex shedding at P02 is higher than the one at P01 due to the impingement of the vortices shed from the upwind box.

5.4. Contribution of local lift coefficient

In this section, the contributions of the local lift coefficient are analysed in order to identify the deck zones contributing to the overall lift, hence enhancing deck oscillations. The total lift force enhancement contribution of each probe is calculated as indicated in Eq. (26), where the correlation of overall and local lift coefficients are multiplied by

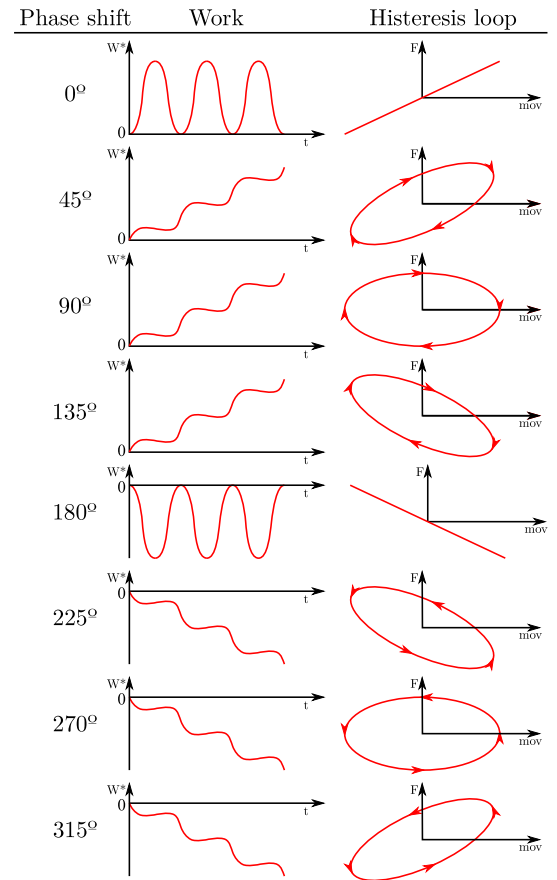


Fig. 14. Relation between phase-shift, cumulative work, loop shape and rotation for hysteresis loops of two pure sinusoidal signals. (W^* is the cumulative work).

the standard deviation of the local lift coefficient (Hu et al., 2022). Positive values indicate lift forcing enhancement while negative values represent locations counteracting lift forcing on the deck. The magnitude indicates the intensity of the local contribution to the overall twin-box deck time-dependent lift forcing, which is not provided by

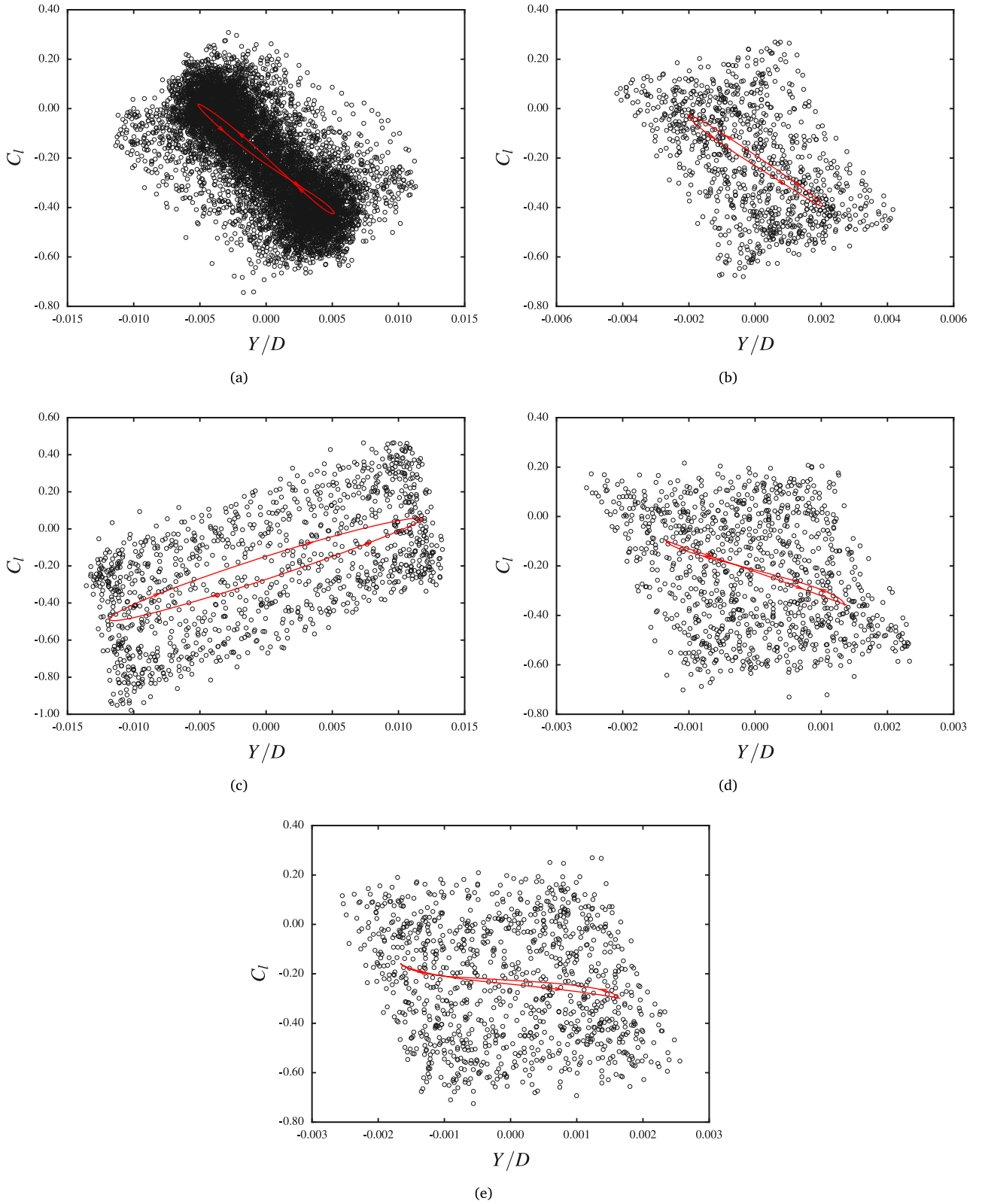


Fig. 15. Phase averaged hysteresis loops of the lift coefficient and heave oscillation. (a) $U/(f_0 B) = 0.3$, (b) $U/(f_0 B) = 0.325$, (c) $U/(f_0 B) = 0.35$, (d) $U/(f_0 B) = 0.375$ and (e) $U/(f_0 B) = 0.4$. The black circles are representing the lift coefficient vs. Y/D points. (Note the different scales in the C_l and Y/D axes for the different figures).

the correlation coefficient (Hu et al., 2022). The spanwise averaged contribution distribution is presented in Fig. 12 for the set of considered reduced velocities, finding that the contribution to time-dependent lift forcing of the upwind box is minimal and only relevant at VIV. This is related with the low values of the standard deviation of the pressure coefficient ($\bar{C}_p$) over the upwind box (see Fig. 6). The larger contribution values are obtained around the downwind box, indicating that VIV excitation is controlled by the downwind box which was also mentioned in Section 5.2 based on the analysis of the spectra. The maximum contribution values are obtained, as expected, for the peak amplitude case ($U/(f_0 B) = 0.35$). However, the contribution values do not consider explicitly the oscillation of the deck, hence they only provide an indirect qualitative assessment of the VIV response.

5.5. Hysteresis loops

In order to further analyse the VIV phenomenon of the Stonecutters bridge bare deck, the trajectories of the lift coefficient with respect to the heave oscillation have been studied. As the lift and heave displacement time-histories present multiple frequency components, the cycles are not regular, presenting changes in orientation and rotation. This can be seen in Fig. 13, where the trajectory over a few cycles of oscillation for $U/(f_0 B) = 0.35$ (peak amplitude) are presented.

The study of hysteresis loops also provides information about the work done by the dynamical system and the phase-lag of the oscillation with respect to the lift force. A chart sketching the possibilities for phase-shift, work and loop shape for two pure sinusoidal signals is presented in Fig. 14 for assisting in the understanding of the phase averaged hysteresis loops presented in Fig. 15, where the cloud of points is representative of the irregular trajectories of the system. The phase-averaging technique is adopted to represent the global trajectory along the studied time length. In Fig. 15, it can be seen that the case corresponding to the maximum peak amplitude, $U/(f_0 B) = 0.35$ (Fig. 15(c)), presents a well defined phase-averaged hysteresis loop, corresponding with a positive phase-shift and with a clockwise rotation, representative of positive work associated with a sustained VIV excitation. For the other cases, the loops are rotating counter-clockwise, indicative of negative work of the system, inhibiting VIV, and hence associated with lower oscillations, as obtained by both the CFD simulations and the wind tunnel tests adopted for validation.

5.6. Work done by the deck during heave oscillation

In order to complement the findings related with the hysteresis loop analysis (Section 5.5), the work produced by the lift force on the complete deck and on each individual box are studied in this section. These analyses make possible to shed additional light on the VIV excitation mechanism, identifying the regions of the deck that are contributing the most to sustain the oscillations.

The cumulative work produced by the overall deck lift along the flow-induced heave oscillation, and by each independent box, are studied in this section. The evolution of the cumulative work of the total lift for the considered reduced velocities is shown in Fig. 16, meanwhile the contributions of each box are presented in Fig. 17. Fig. 16 depicts how positive work builds up at $U/(f_0 B) = 0.3$ and 0.35 , the reduced velocities at which the amplitudes of heave oscillation show larger values (see Fig. 7). The presence of regions in Fig. 16 where the cumulative work keeps relatively constant, for example the region in the interval $tU/D = (600-700)$ for a reduced velocity of $U/(f_0 B) = 0.30$, is related with the time-histories of the lift coefficient and heave-oscillation being at anti-phase (see Fig. 14), and the small oscillations in the cumulative work in that time range are due to the low amplitude of the heave oscillations.

Fig. 17 provides a good explanation about the VIV excitation mechanism in this twin-box deck. It is noticeable that the upwind box presents positive work (Fig. 17(a)) for almost all the considered reduced velocities, particularly for $U/(f_0 B) = 0.30$, at which moderate heave

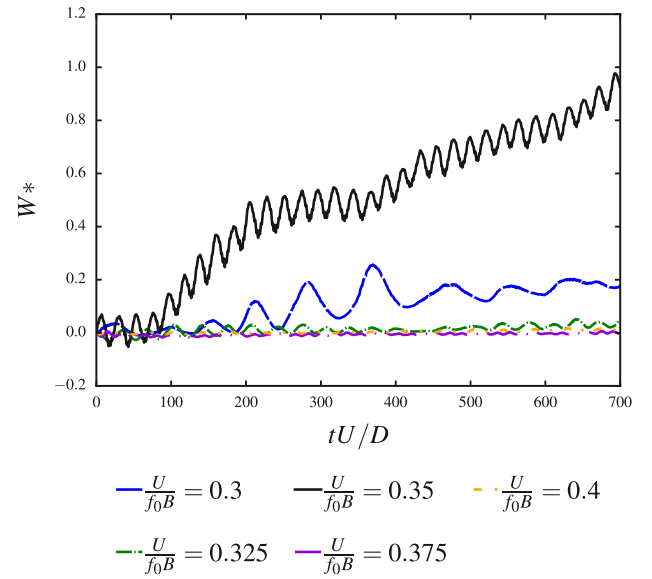


Fig. 16. Evolution of the non-dimensional cumulative work by the total lift and heave oscillation for all the considered reduced velocities.

oscillation was obtained (Fig. 7). However, it is remarkable that for the peak heave oscillation reduced velocity ($U/(f_0 B) = 0.35$), the cumulative work at the upwind box is negative.

It is noted that the contribution of the local lift (Section 5.4) does not explain by itself the VIV excitation process, as the oscillation of the system is not explicitly considered. For example, for the peak response case ($U/(f_0 B) = 0.35$), the high contribution in the upwind box is associated with negative cumulative work. On the other hand, Fig. 17(b), reporting the cumulative work done by the downwind box, presents a very different scenario. For the peak heave oscillation ($U/(f_0 B) = 0.35$) the work is positive, and so intense that it compensates the negative work done by the upwind box, producing an overall positive cumulative work (Fig. 16), in agreement with the peak response obtained at that reduced velocity. For the other reduced velocities, the cumulative work is negative, counteracting the positive works generated by the windward box. It is noted that for $U/(f_0 B) = 0.30$ the overall work is still positive, providing the moderate heave oscillation mentioned above (Fig. 7). For the other reduced velocities, the negative work done by the downwind box balances the positive work done by the downwind box, providing an almost null overall work, in agreement with the negligible heave oscillation obtained in the CFD simulations and wind tunnel tests for $U/(f_0 B) = 0.325, 0.375$ and 0.40 . In summary, for a given deck geometry, depending on the flow speed, the downwind box prevents or enables the VIV excitation of the deck.

In Fig. 18, the spanwise-averaged time-averaged work distribution over the surface of the deck is reported, assisting in identifying the zones in the boxes contributing to VIV excitation. It can be noticed that the distributions are very similar among the cases showing minor VIV excitation and consequently almost null work. The spanwise-averaged time-averaged work is only relevant in the upwind box in the regions close to the gap. Over the downwind box, again the work is almost null, with the exception of two regions of positive and negative work on the upper surface, close to the gap, providing net negative work as per Fig. 17(b).

Major differences in the distributions correspond with $U/(f_0 B) = 0.35$, as the positive work for the peak amplitude case is mainly done by the downwind box with negative net work on the upwind box. Positive work around the leeward box is generated in the trailing half of the upper and lower panels, in the inclined panel facing the gap, and in the windward region of the upper panel, where vortices shed from the upwind box are impinging upon. This distribution suggest the potential

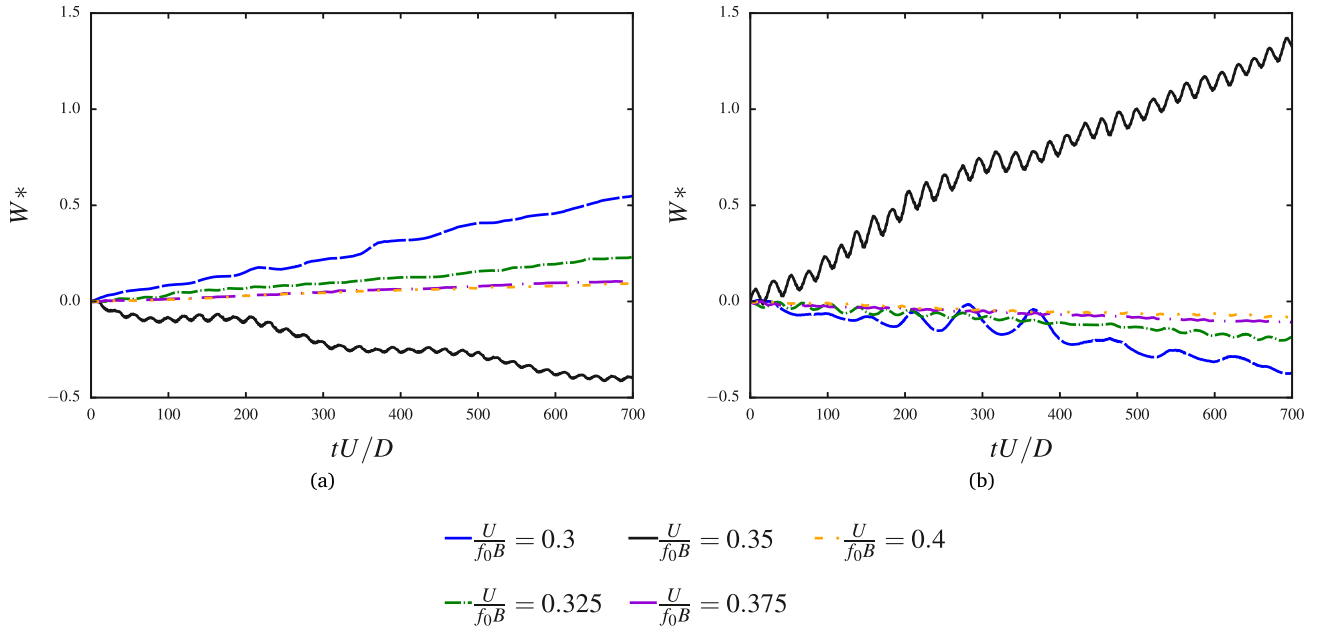


Fig. 17. Evolution of the non-dimensional cumulative work by the lift of each box and heave oscillation for all the considered reduced velocities: (a) upwind box and (b) downwind box .

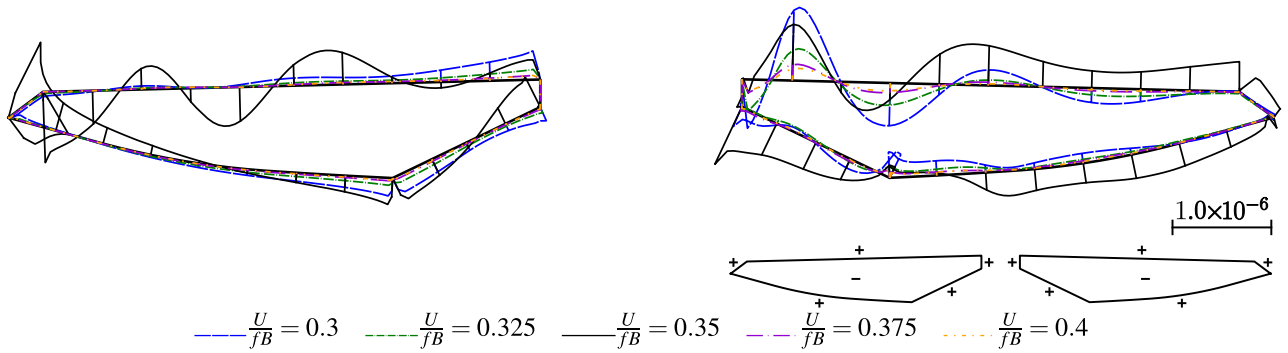


Fig. 18. Spanwise-averaged time-averaged work distribution.

for VIV mitigation if the work at the inclined and upper panels of the downwind box facing the gap is reduced. The installation of guide vanes in the Stonecutters Bridge, which were successful mitigating VIV by suppressing the vortex shedding from the upwind box, represents the engineering-based approach to the reduction of the spanwise-averaged time-averaged work in the critical regions of the leeward box.

These results show the key role played by the leeward box in the VIV response of twin-box decks, as it controls the overall cumulative work done by the deck. Furthermore this sheds additional light about the importance of the gap distance between box girders, as this distance governs the vortex shedding frequency and the phase-lags between the flow-induced forces acting on each individual box (Álvarez et al., 2018). Furthermore, mitigation strategies aiming at preventing VIV of twin-box deck should focus on modifying the flow patterns and/or forcing-oscillation coupling at the leeward box.

6. Conclusions

The spatial and temporal discretisation sensitivity studies based on the assessment of the uncertainty in the integral parameters have shown that, for the considered application case, the medium mesh,

featuring 3.2 million cells, along with a maximum Courant number of 1.0, strikes the right balance between accuracy and computational burden. The decrease in the ratio of modelled turbulence to total turbulence in the fine mesh (8.6 million cells) with respect to the medium mesh did not significantly impact the values obtained for the integral parameters. The sensitivity studies reported herein suggest that maximum uncertainties in the force coefficients, non-dimensionalised with the single box width C as reference length, in the order of $1 \cdot 10^{-2}$ represent a safe reference value for the selection of the mesh refinement level in 3D LES simulations of this kind.

The numerical results obtained for integral parameters and mean and fluctuating pressure coefficients were in agreement with equivalent experimental data. The values obtained for validation metrics of general use in the discipline have confirmed this good alignment, complementing the engineering judgement with objective measures for accuracy assessment.

The 3D LES simulations under free heave oscillation have successfully reproduced the VIV response over the considered range of reduced velocities. The dominant role played by the downwind box in the VIV process starts to emerge as the total lift force time-history is dominated by the lift force acting on the leeward deck.

The analysis of the spectra of time-histories of the lift force acting on the overall deck and each individual box, along with heave oscillation, has shed light on the complex interplay between the aerodynamic forcing and the dynamic response of the system. It is to note that the reduced velocity $U/(f_0 B) = 0.30$ shows a moderate level of heave oscillation at the vortex shedding frequency, which is associated with the lift forcing at the same frequency. For the reduced velocity $U/(f_0 B) = 0.35$, at partial lock-in, the amplitude of oscillation is the highest, with vortices being shed from the windward box having relevant energy content at both the natural and vortex shedding frequencies, impinging on the leeward box, whose lift coefficient spectrum features two peaks of similar intensity at both frequencies, hence preventing the full lock-in of the deck. Further studies may be required for a more detailed description of the excitation mechanism at VIV; for instance, addressing the possibility of pressure perturbations being fed back upstream (Naudascher and Rockwell, 1994) due to the flow interaction with the downstream box, or the role played by the box geometry facing the gap between girders.

Similarly, the information provided by the spectra of the pressure coefficients at selected positions on the deck has shown the importance of the aerodynamic loads acting on the downwind box surfaces in the vicinity of the gap for the VIV excitation, providing a sound support for the general accepted knowledge about the importance of the gap distance in this bridge typology.

The analysis of the span-averaged local lift coefficient contribution distributions has shown very different patterns in the windward and leeward boxes. The contribution is minor on the upwind box, and only relevant at partial lock-in ($U/(f_0 B) = 0.35$); nevertheless, high contributions have been obtained around the downwind box, specially over the regions in the vicinity of the gap between boxes, pointing also to the important role played by the downwind box in the VIV process.

A deeper understanding of the VIV mechanism in twin-box bridges requires taking into consideration the relationship between aerodynamic forcing and heave oscillation. This is achieved by considering the work exerted by the deck, but most remarkably, the work done by each individual box and its distribution around the deck. In this piece of research it has been demonstrated that the VIV response of the deck is governed by the work done by the downwind box, which is of different sign from the one exerted by the upwind box. At partial lock-in, the positive work in the leeward box exceeds the negative one from the windward box, resulting in sustained heave oscillations. For $U/(f_0 B) = 0.30$, the negative work done by the leeward box is not enough to fully compensate the positive work in the windward box,

resulting in a slight heave excitation. For the remaining studied reduced velocities both works in the windward and leeward boxes compensate, resulting in negligible heave oscillations. Interestingly, the distribution of the work around the deck, allows the identification of the regions in the deck where positive work takes place, enabling the design of aerodynamic countermeasures tailored for the specific twin-box deck geometry under consideration.

The potential of CFD-based approaches for the detailed study of the complex fluid–structure interaction at VIV of twin-box decks must be emphasised. Here, the CFD simulations, validated with wind tunnel data, have enabled an in-depth analysis of the aerodynamic and heave oscillation responses at different reduced velocities, considering an ample range of output data. This points towards the gradual unlocking of data-driven techniques in the aeroelastic design of long-span bridges.

For this twin-box deck configuration, it has been demonstrated the different behaviour of the windward and leeward boxes at several reduced velocities, and how it is the downwind box the one responsible for the overall deck VIV response as it enables or prevents the vortex-induced excitation of the deck by the partial transfer of energy in the lift force spectrum from the vortex shedding frequency to the natural frequency of the system.

CRediT authorship contribution statement

A.J. Álvarez: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **F. Nieto:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The financial support provided by project PID2023-153061OB-I00 funded by MICIU/AEI 10.13039/501100011033 and by FEDER, EU. Also the support of projects ED481B-2021-133 and ED431C 2021/33 of the Galician Regional Government is acknowledged. The computations have been carried out in the Galicia Supercomputing Centre (CESGA).

Appendix A. Verification data

Table A.1

Spatial discretisation uncertainty. (N is the overall number of elements, h is the characteristic cell size, r is the refinement ratio, p is the apparent order of the method, ϕ is the value of the variable of interest, ϕ_{ext} is the extrapolated value, e_a is the approximate relative error, e_{ext} is the extrapolated relative error, GCI is the grid convergence index, 1,2,3 refers to the fine, medium and coarse mesh respectively, and U_{mesh}^{ij} is the uncertainty value.)

	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	$\widetilde{C_d}$	$\widetilde{C_l}$	$\widetilde{C_m}$	St
N_1, N_2, N_3	8598192, 3218112, 1970208						
h_1, h_2, h_3	$4.881 \times 10^{-03}, 5.198 \times 10^{-03}, 6.773 \times 10^{-03}$						
r_{21}	1.388						
r_{32}	1.178						
ϕ_1	0.151	-0.263	0.226	0.023	0.166	0.076	0.247
ϕ_2	0.153	-0.260	0.206	0.027	0.202	0.090	0.244
ϕ_3	0.154	-0.244	0.183	0.033	0.244	0.118	0.235
p	2.990	11.661	3.507	4.653	3.588	6.190	7.858
ϕ_{ext}^{21}	0.150	-0.263	0.236	0.021	0.149	0.074	0.248
ϕ_{ext}^{32}	0.150	-0.263	0.235	0.021	0.149	0.074	0.248
e_a^{21}	1.138%	1.047%	8.999%	18.059%	21.959%	18.417%	1.333%
e_a^{32}	1.136%	6.202%	11.215%	22.301%	20.791%	31.380%	3.824%
e_{ext}^{21}	0.689%	0.023%	4.010%	5.295%	10.874%	2.872%	0.110%
e_{ext}^{32}	1.835%	1.070%	12.649%	24.311%	35.221%	21.819%	1.441%
GCI_{fine}^{21}	0.856%	0.029%	5.222%	6.286%	12.259%	3.490%	0.137%
GCI_{medium}^{32}	2.253%	1.352%	18.100%	24.446%	32.559%	22.389%	1.828%
GCI_{coarse}	9.781%	415.186%	101.314%	240.230%	189.642%	468.025%	86.679%
U_{fine}^{21}	1.292×10^{-03}	7.701×10^{-05}	1.180×10^{-02}	1.417×10^{-03}	2.030×10^{-02}	2.658×10^{-03}	3.398×10^{-04}
U_{medium}^{32}	3.441×10^{-03}	3.512×10^{-03}	3.722×10^{-02}	6.506×10^{-03}	6.576×10^{-02}	2.019×10^{-02}	4.459×10^{-03}
U_{coarse}	1.511×10^{-02}	1.012	1.849×10^{-01}	7.819×10^{-02}	4.627×10^{-01}	5.546×10^{-01}	2.034×10^{-01}
Oscillatory convergence	No	No	No	No	No	No	No

Table A.2

Temporal discretisation uncertainty. (h is the Courant number, r is the refinement ratio, p is the apparent order of the method, ϕ is the value of the variable of interest, ϕ_{ext} is the extrapolated value, e_a is the approximate relative error, e_{Ext} is the extrapolated relative error, GCI is the grid convergence index, 1,2,3 refers to short, regular and large time steps respectively, and $U_{Timestep}^{ij}$ is the uncertainty value.)

	$\overline{C_d}$	$\overline{C_l}$	$\overline{C_m}$	$\widetilde{C_d}$	$\widetilde{C_l}$	$\widetilde{C_m}$	St
h_1, h_2, h_3	0.5, 1.0, 2.5						
r_{21}	2.000						
r_{32}	2.500						
ϕ_1	0.153	-0.260	0.202	0.028	0.221	0.095	0.246
ϕ_2	0.153	-0.260	0.206	0.027	0.202	0.090	0.244
ϕ_3	0.153	-0.265	0.203	0.026	0.197	0.088	0.241
p	2.008	3.750	0.700	3.058	2.850	2.319	0.048
ϕ_{ext}^{21}	0.153	-0.260	0.196	0.029	0.224	0.097	0.299
ϕ_{ext}^{32}	0.153	-0.260	0.209	0.027	0.202	0.090	0.299
e_a^{21}	0.056%	0.062%	1.927%	6.094%	8.679%	5.315%	0.732%
e_a^{32}	0.397%	2.012%	1.285%	1.646%	2.679%	2.079%	1.013%
e_{ext}^{21}	0.019%	0.005%	3.183%	0.825%	1.379%	1.315%	17.706%
e_{ext}^{32}	0.075%	0.067%	1.409%	1.062%	0.212%	0.028%	18.308%
GCI_{short}^{21}	0.023%	0.006%	3.856%	1.040%	1.748%	1.666%	26.894%
$GCI_{regular}^{32}$	0.094%	0.084%	1.787%	0.133%	0.266%	0.353%	28.013%
GCI_{large}	2.371%	34.964%	5.513%	18.239%	26.051%	14.724%	30.275%
U_{short}^{21}	3.558×10^{-05}	1.616×10^{-05}	7.779×10^{-03}	2.946×10^{-04}	3.865×10^{-03}	1.587×10^{-03}	6.609×10^{-02}
$U_{regular}^{32}$	1.431×10^{-04}	2.174×10^{-04}	3.674×10^{-03}	3.538×10^{-05}	5.363×10^{-04}	3.180×10^{-04}	6.834×10^{-02}
U_{large}	3.636×10^{-03}	9.266×10^{-02}	1.119×10^{-02}	4.774×10^{-03}	5.121×10^{-02}	1.300×10^{-02}	7.311×10^{-02}
Oscillatory convergence	No	No	Yes	No	No	No	No

Appendix B. Main flow features over one cycle of VIV oscillation at $U/(f_0 B) = 0.35$

Focusing on the upper and lower trailing edges of the upwind box, it is appreciated how the shedding of the vortices is in phase with the heave oscillation. In other words, the vortex builds up at the upper trailing edge along the upward displacement and it is shed when the maximum vertical displacement is reached (P1), meanwhile the vortex formed at the lower trailing edge is shed when the deck reaches the lowest heaving position.

In the downwind box, the formation of vortices due to the deck motion interacts with the impingement of the vortices shed from the upwind box. This is depicted in the 3D plot of Fig. B.2, where neat vortical structures spanning the whole span length are generated at the upwind box leeward corner, meanwhile more disorganised and three dimensional structures are formed around the downwind box. Despite this, it may be noticed also how two vortices spanning the whole span length of the downwind box at the upper leading edge (A) and upper trailing edge (B) are shed at P1, meanwhile at the minimum heaving position (P5), the same process is observed at the lower leading edge (C) and lower trailing edge (D) (see Fig. B.2). Three-dimensional structures like horseshoe vortices may be identified in the gap between boxes.

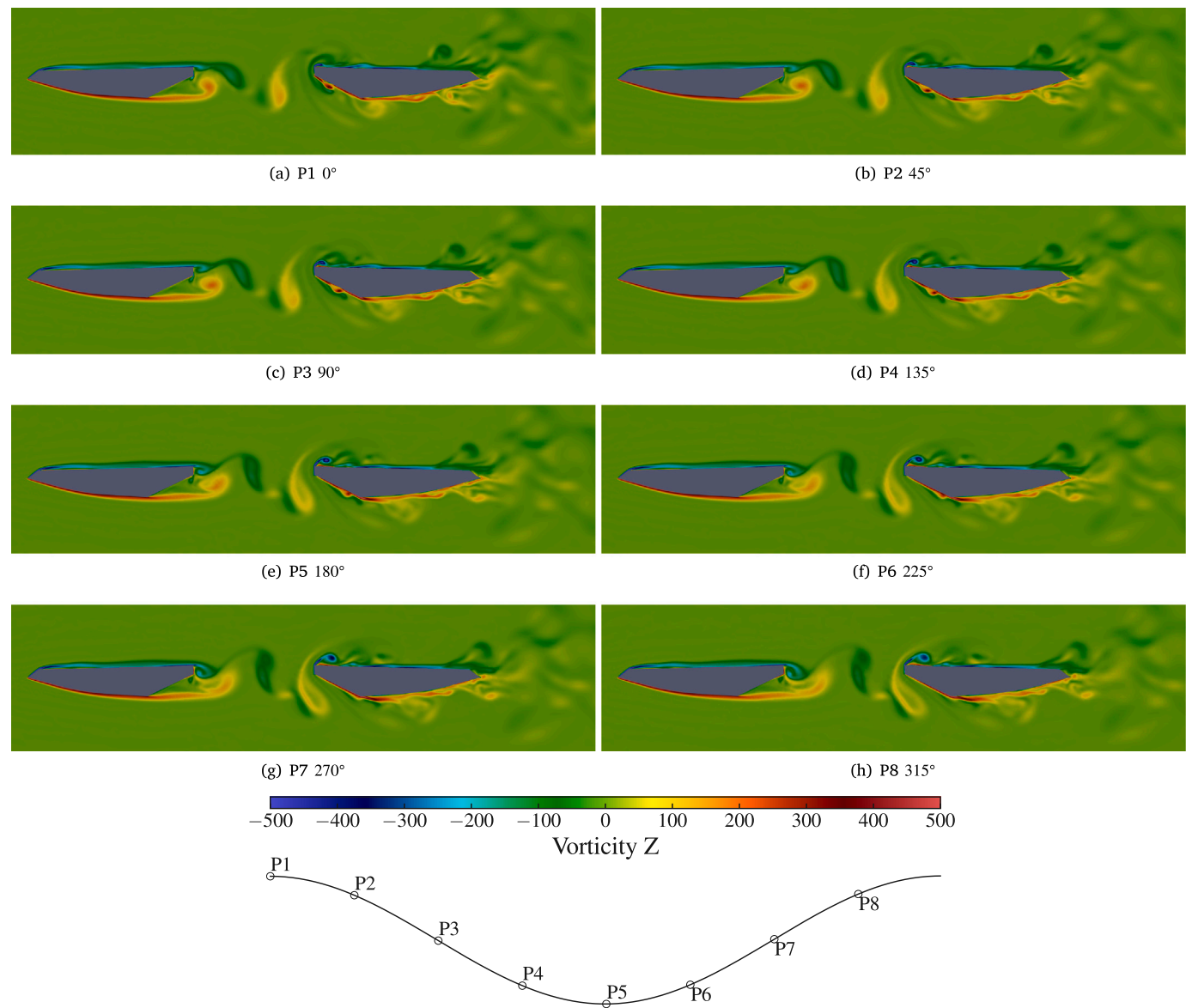


Fig. B.1. Instantaneous vorticity fields over a cycle of oscillation. It is coloured with the Vorticity Z values, positive values correspond with a counter-clockwise rotation around the spanwise axis.

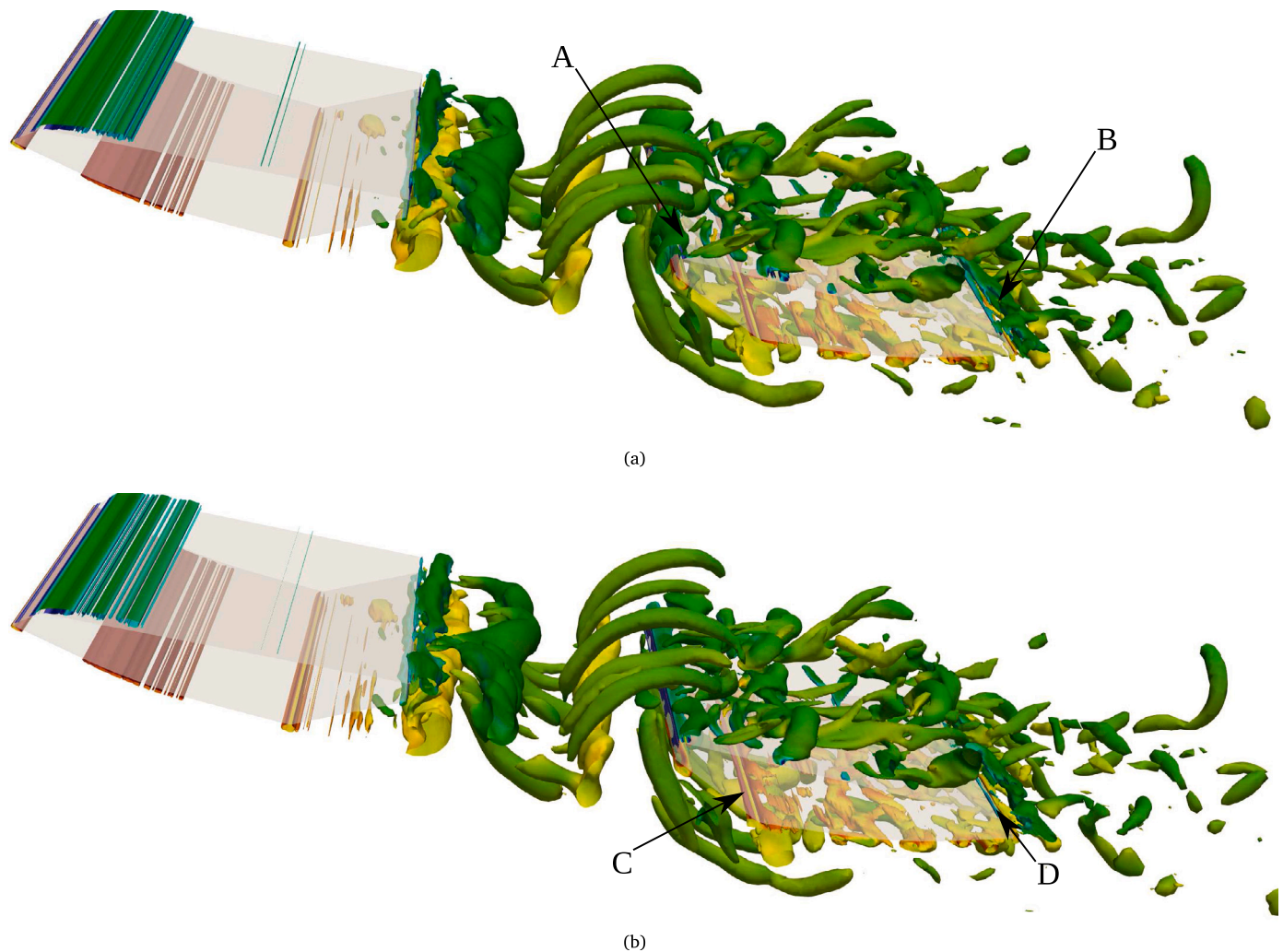


Fig. B.2. Instantaneous Q fields at (a) maximum and (b) minimum heave amplitude oscillation. It is coloured with the Vorticity Z values, positive values correspond with a counter-clockwise rotation around the spanwise axis.

Data availability

Data will be made available on request.

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